

BFS Traversal

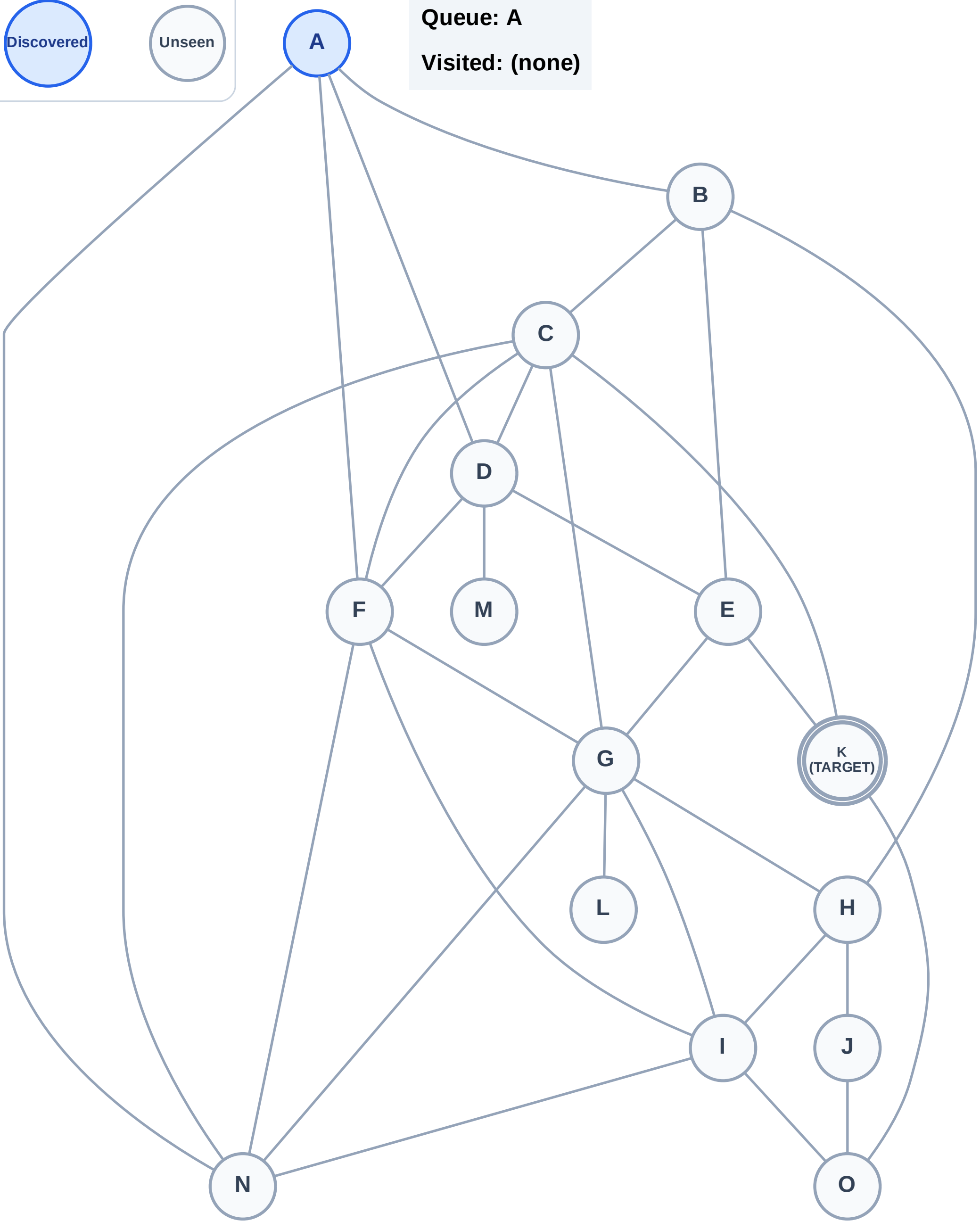
Step 1: Start at A (Seeking target node K)

Legend

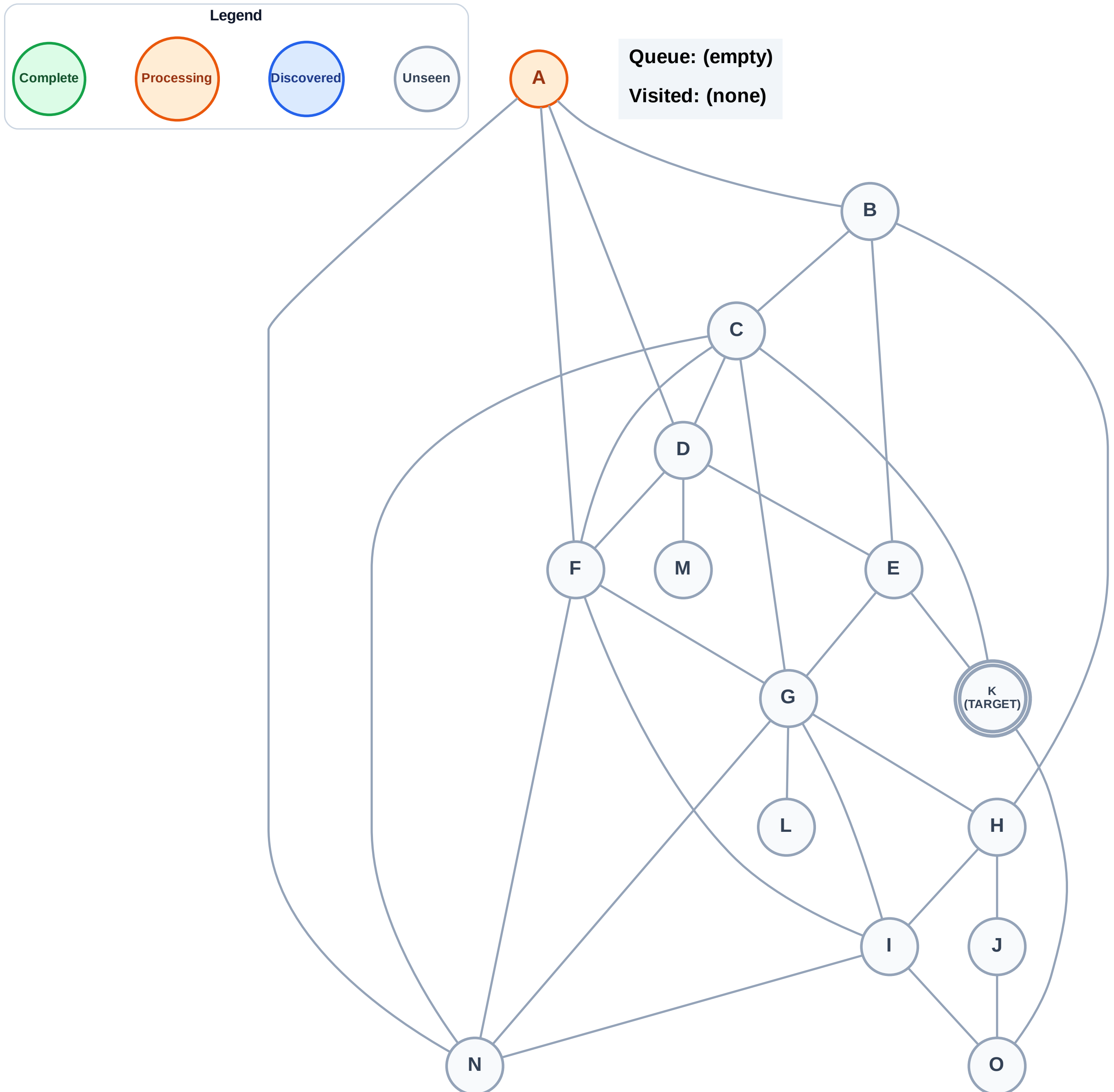


Queue: A

Visited: (none)



Step 2: Process node A



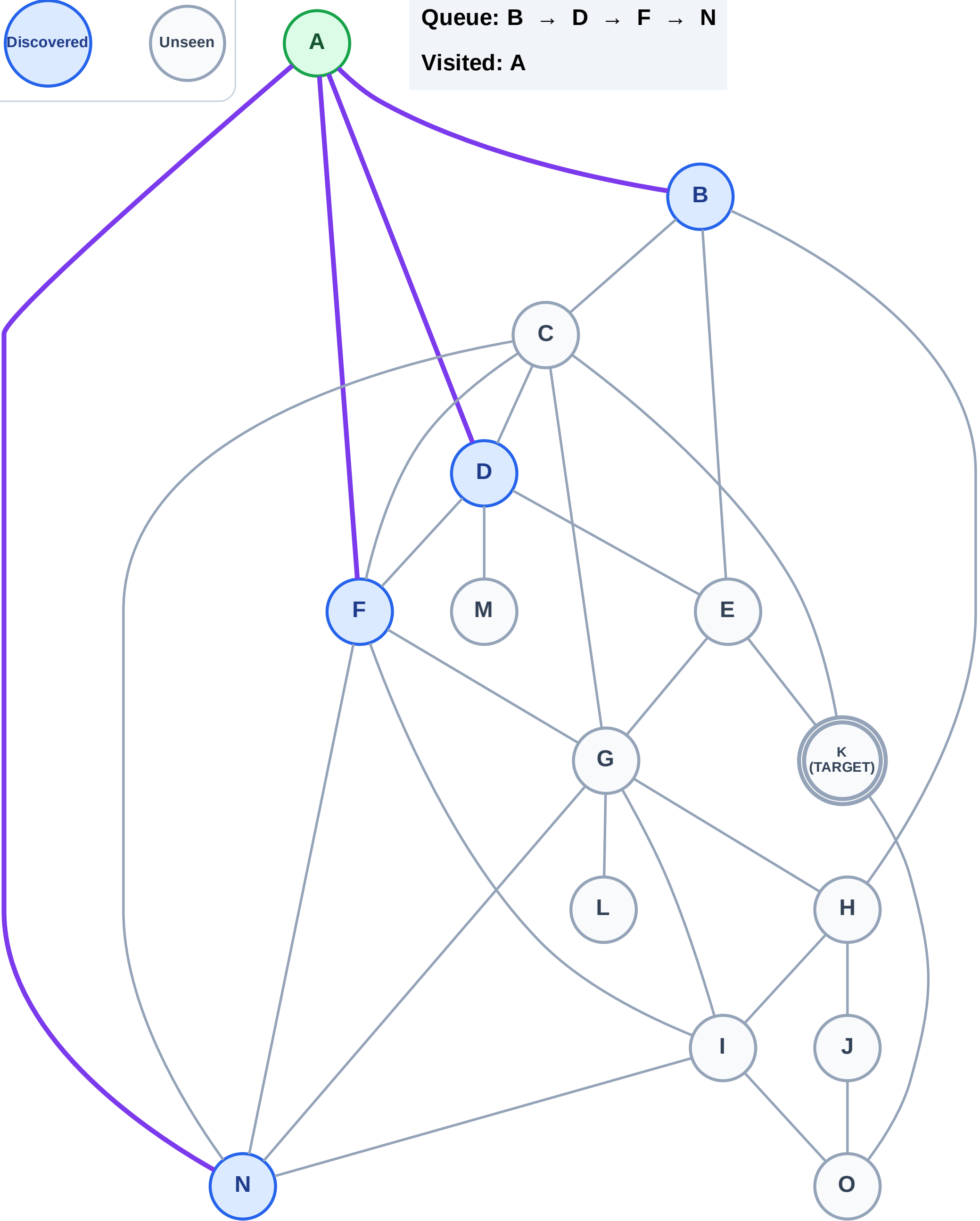
BFS Traversal

Step 3: Discover B, D, F, N from A

Legend



Queue: B → D → F → N
Visited: A



BFS Traversal

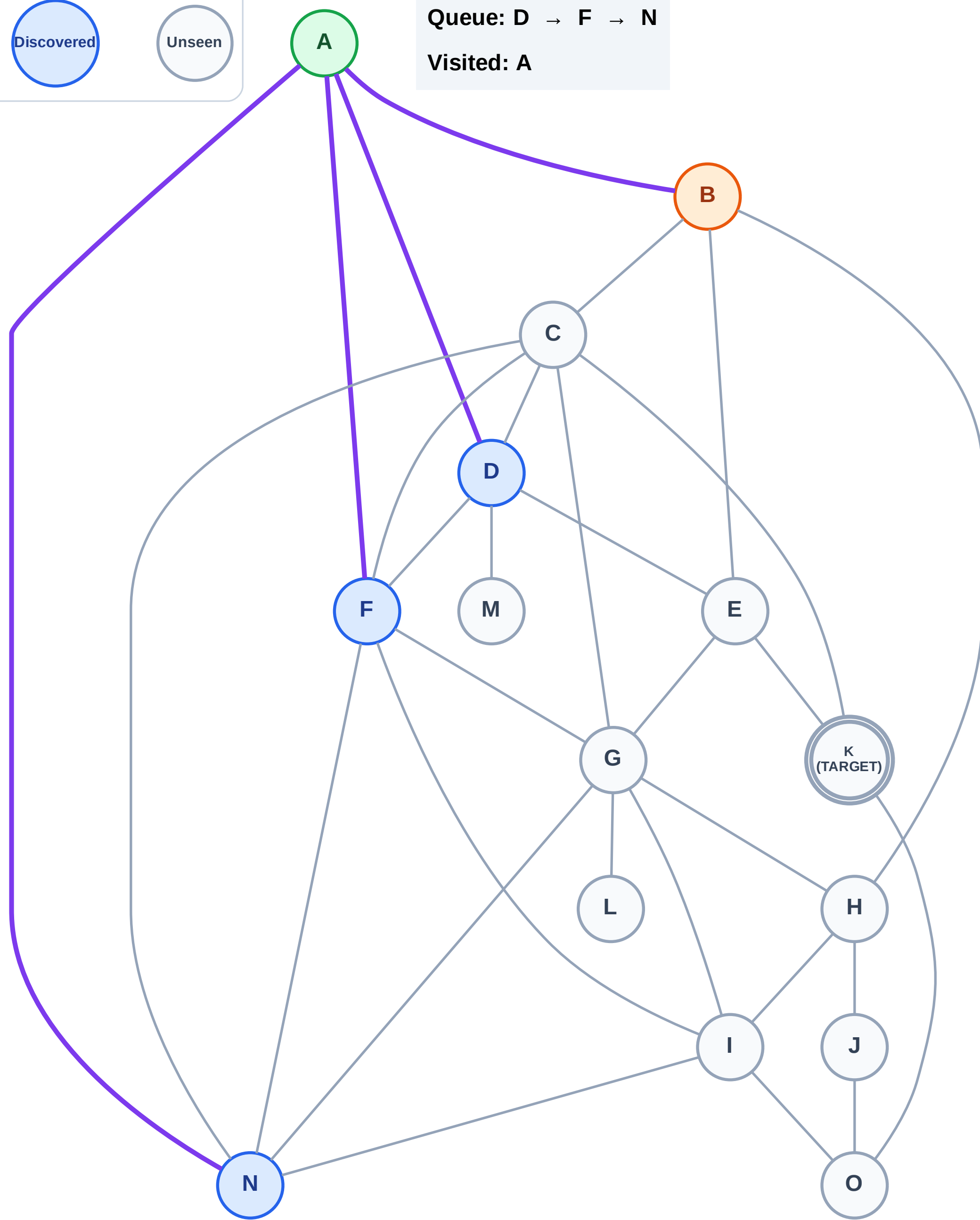
Step 4: Process node B

Legend



Queue: D → F → N

Visited: A

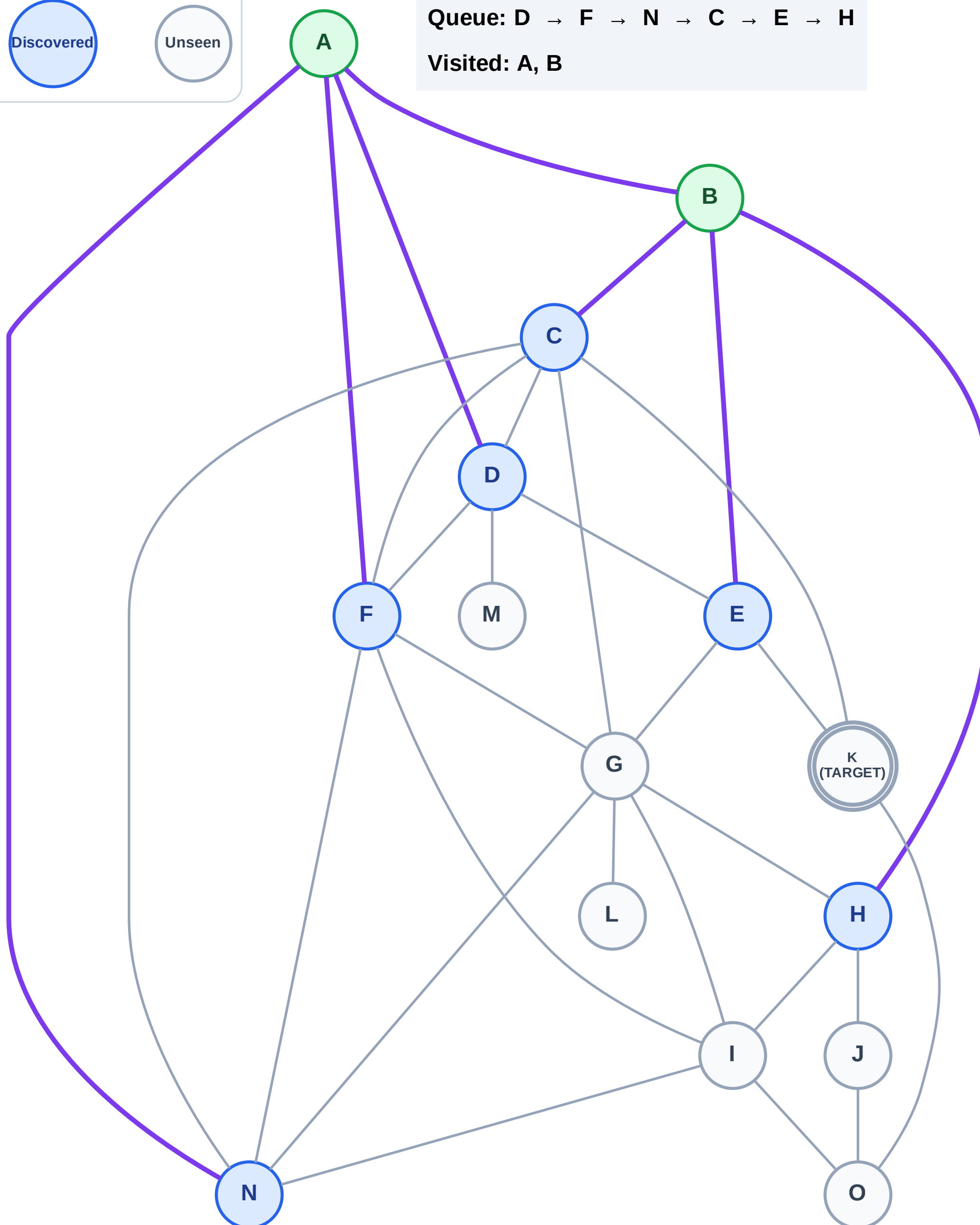


Step 5: Discover C, E, H from B

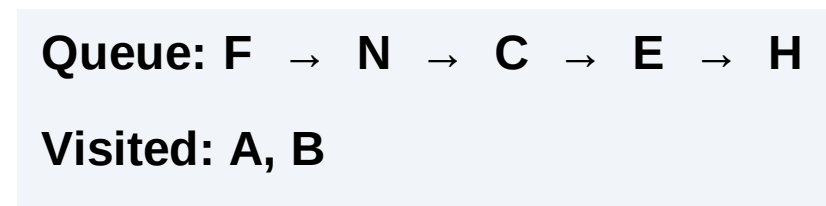
Legend

Unseen

Visited: A, B



Step 6: Process node D



BFS Traversal

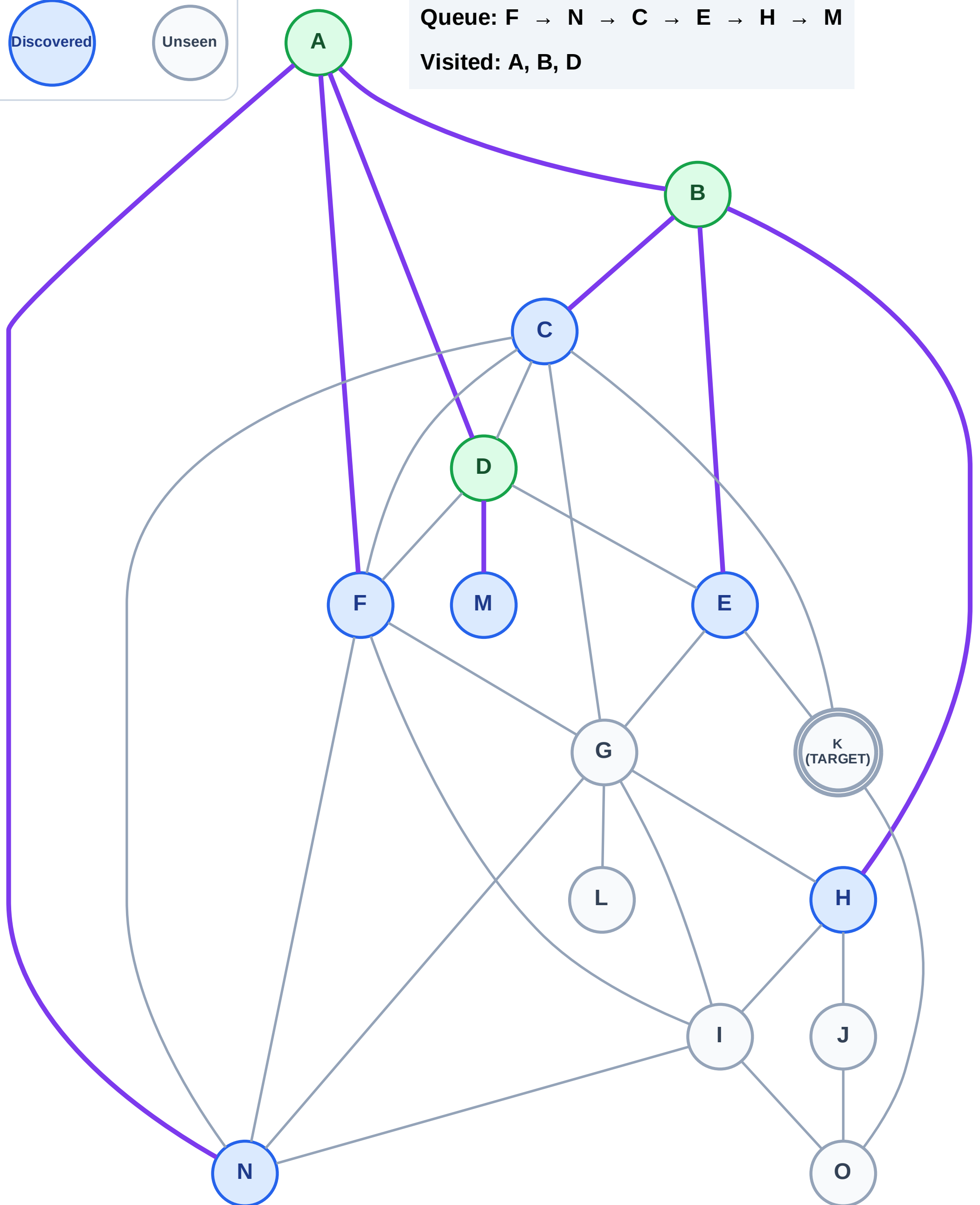
Step 7: Discover M from D

Legend

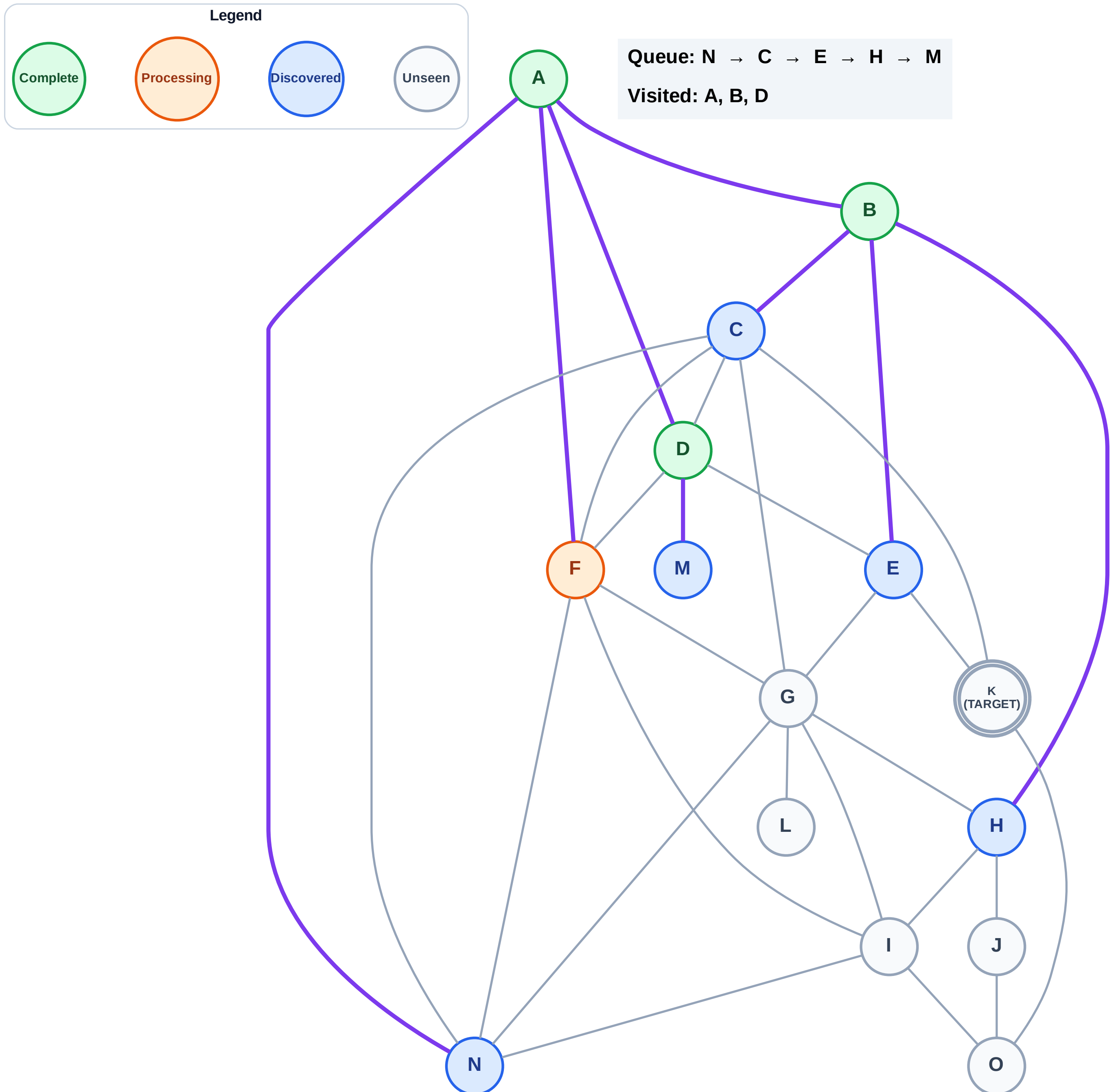


Queue: F → N → C → E → H → M

Visited: A, B, D



Step 8: Process node F



BFS Traversal

Step 9: Discover G, I from F

Legend

Complete

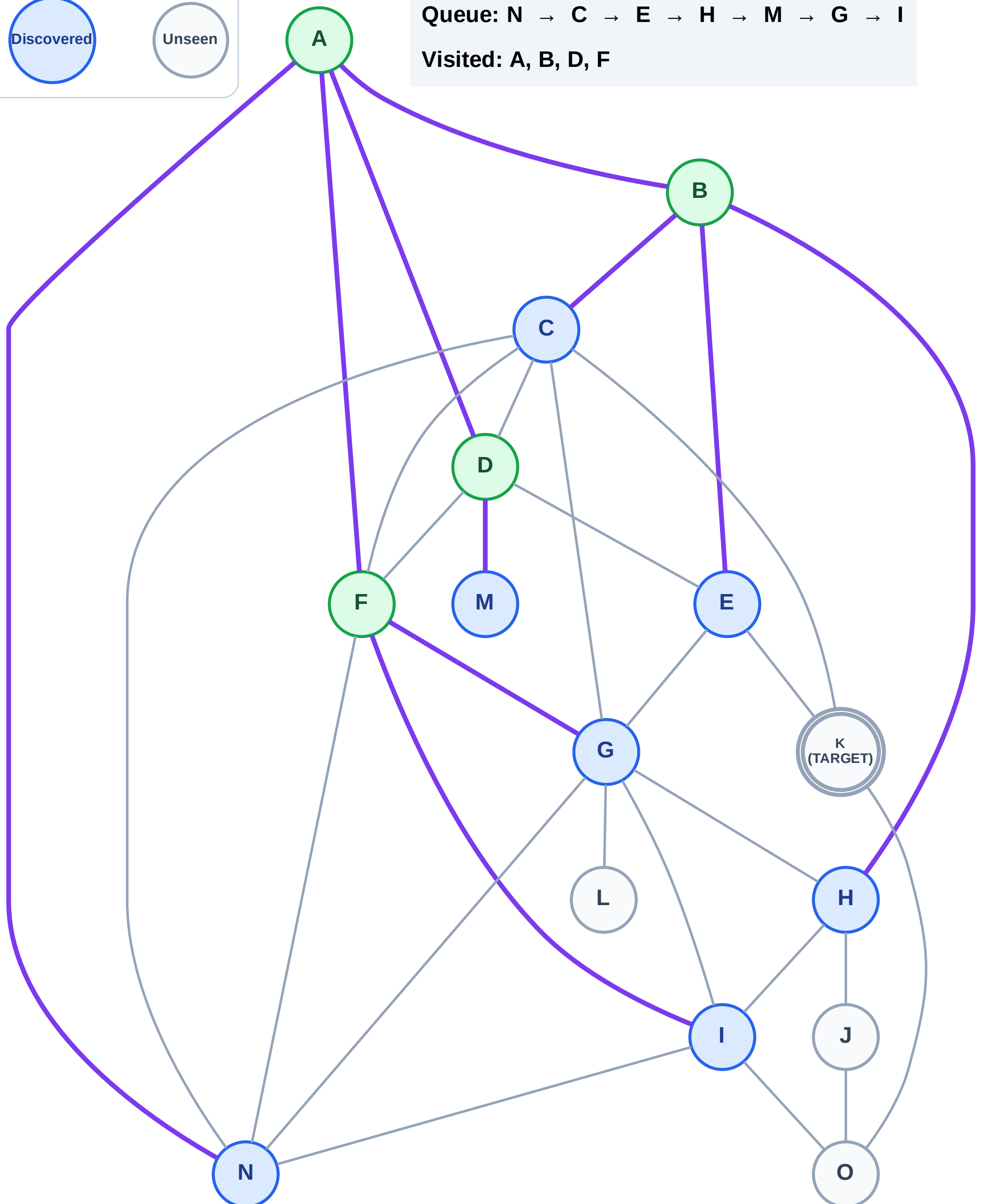
Processing

Discovered

Unseen

Queue: N → C → E → H → M → G → I

Visited: A, B, D, F



BFS Traversal

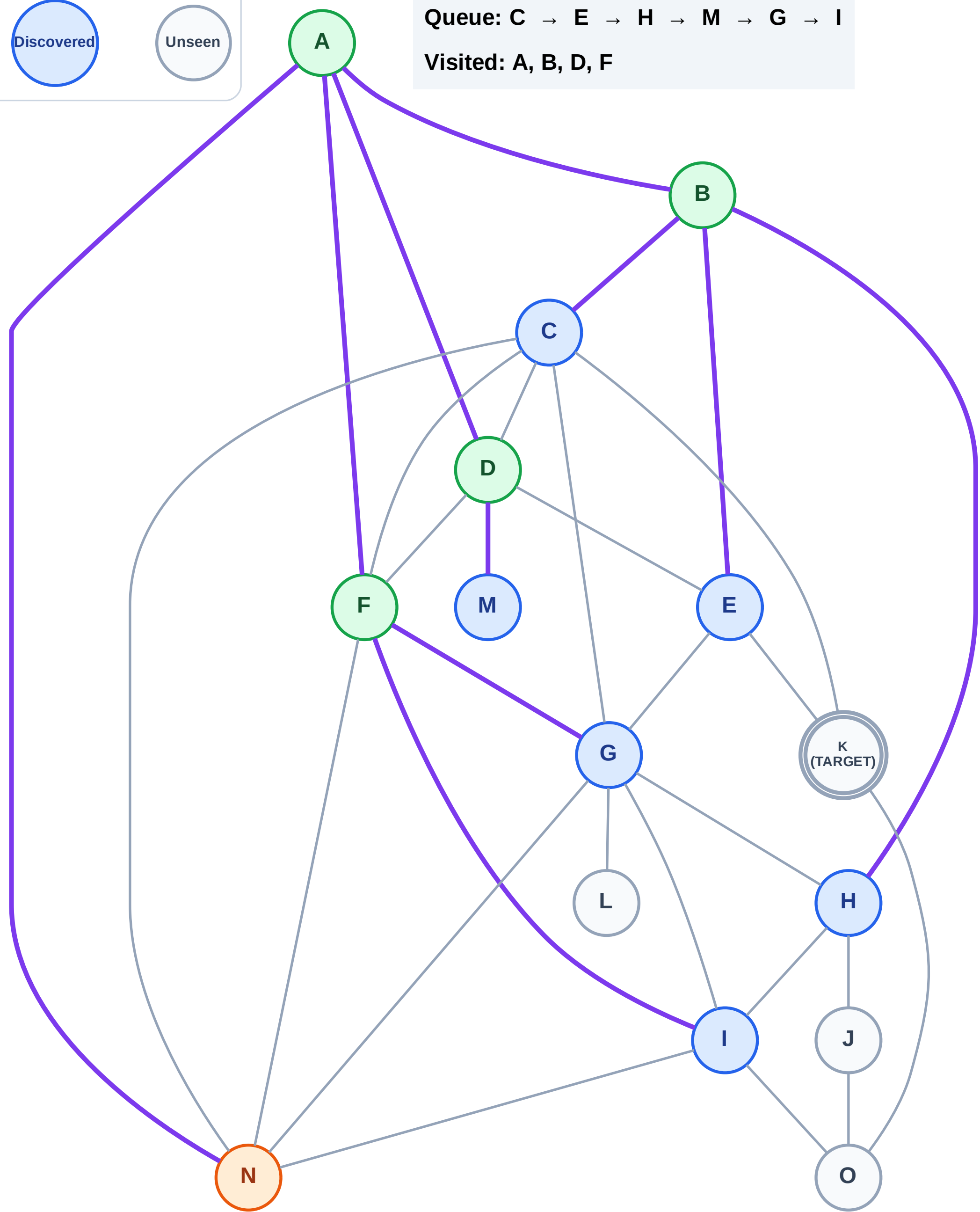
Step 10: Process node N

Legend



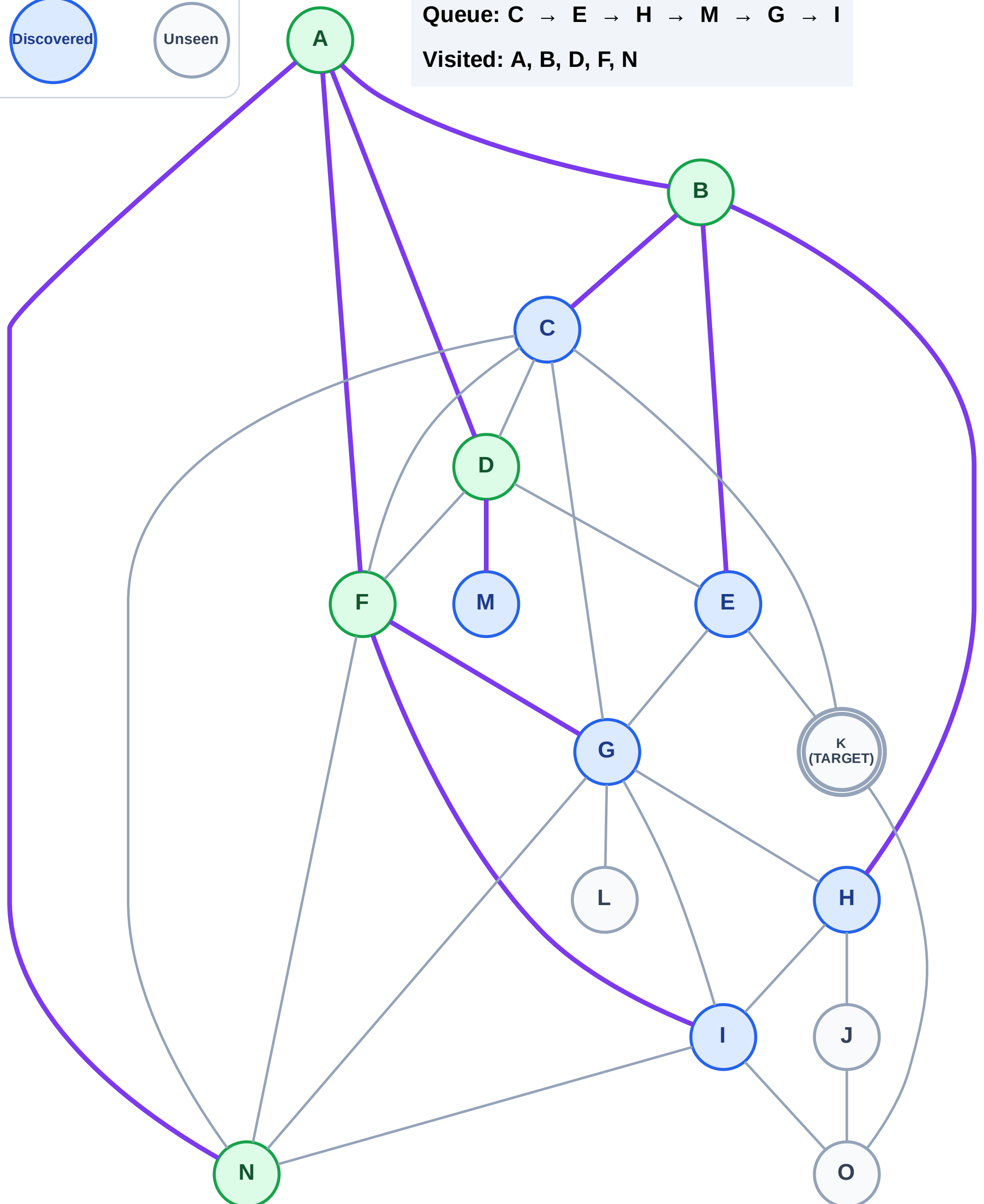
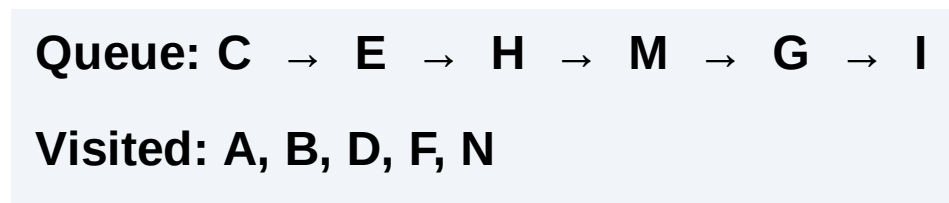
Queue: C → E → H → M → G → I

Visited: A, B, D, F



Step 11: No new neighbors from N

Step 11: No new neighbors from N



BFS Traversal

Step 12: Process node C

Legend

Complete

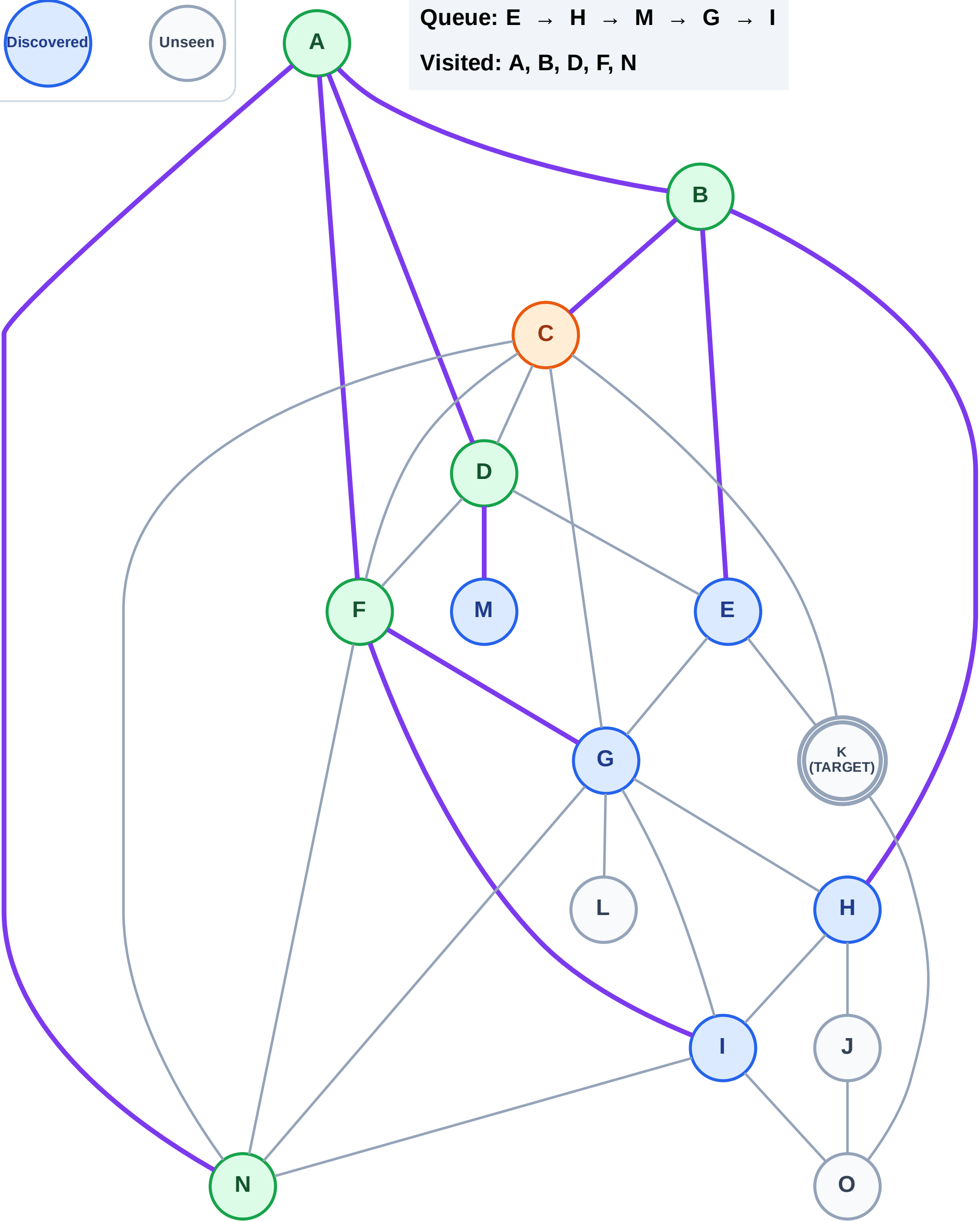
Processing

Discovered

Unseen

Queue: E → H → M → G → I

Visited: A, B, D, F, N



BFS Traversal

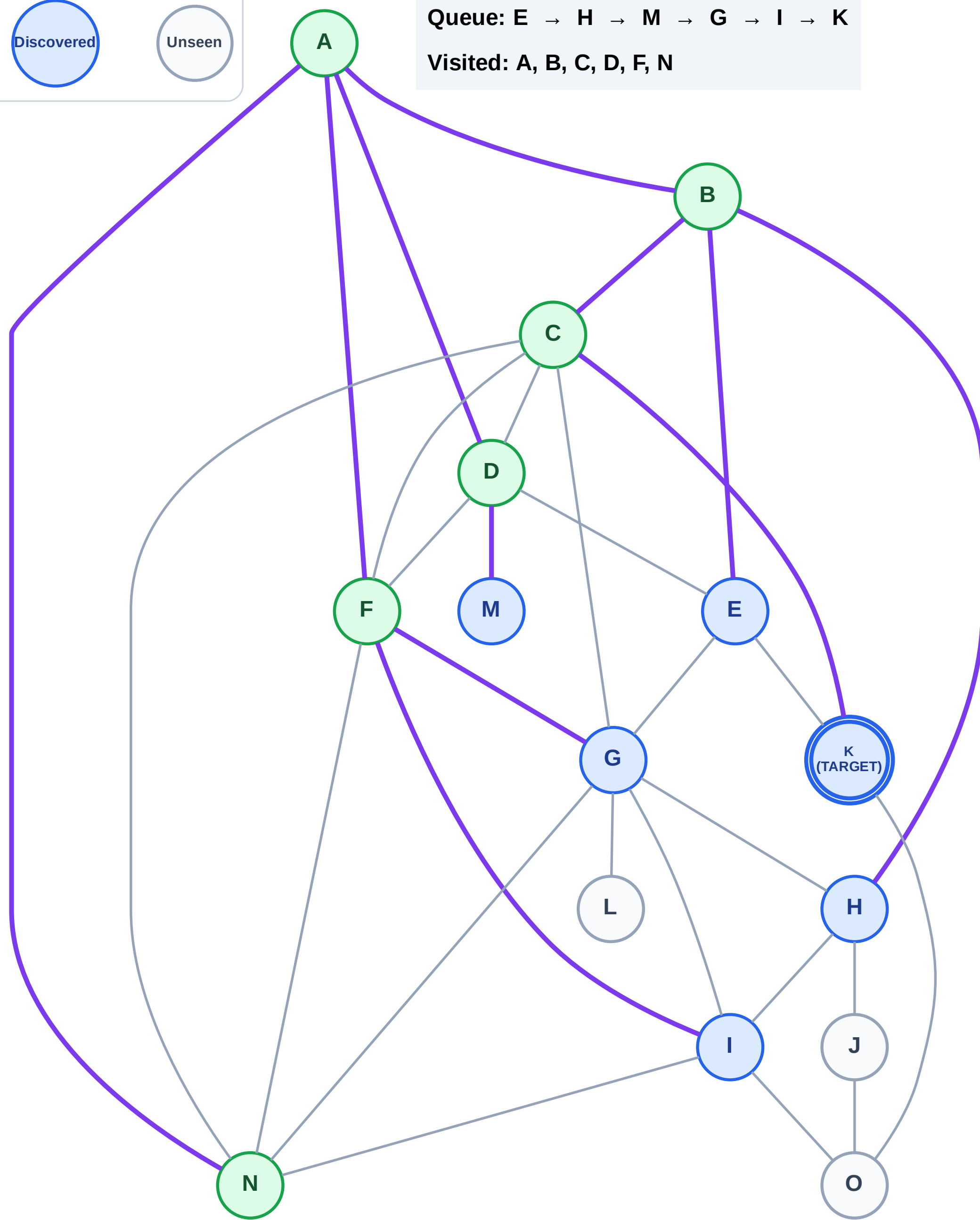
Step 13: Discover K from C

Legend



Queue: E → H → M → G → I → K

Visited: A, B, C, D, F, N



BFS Traversal

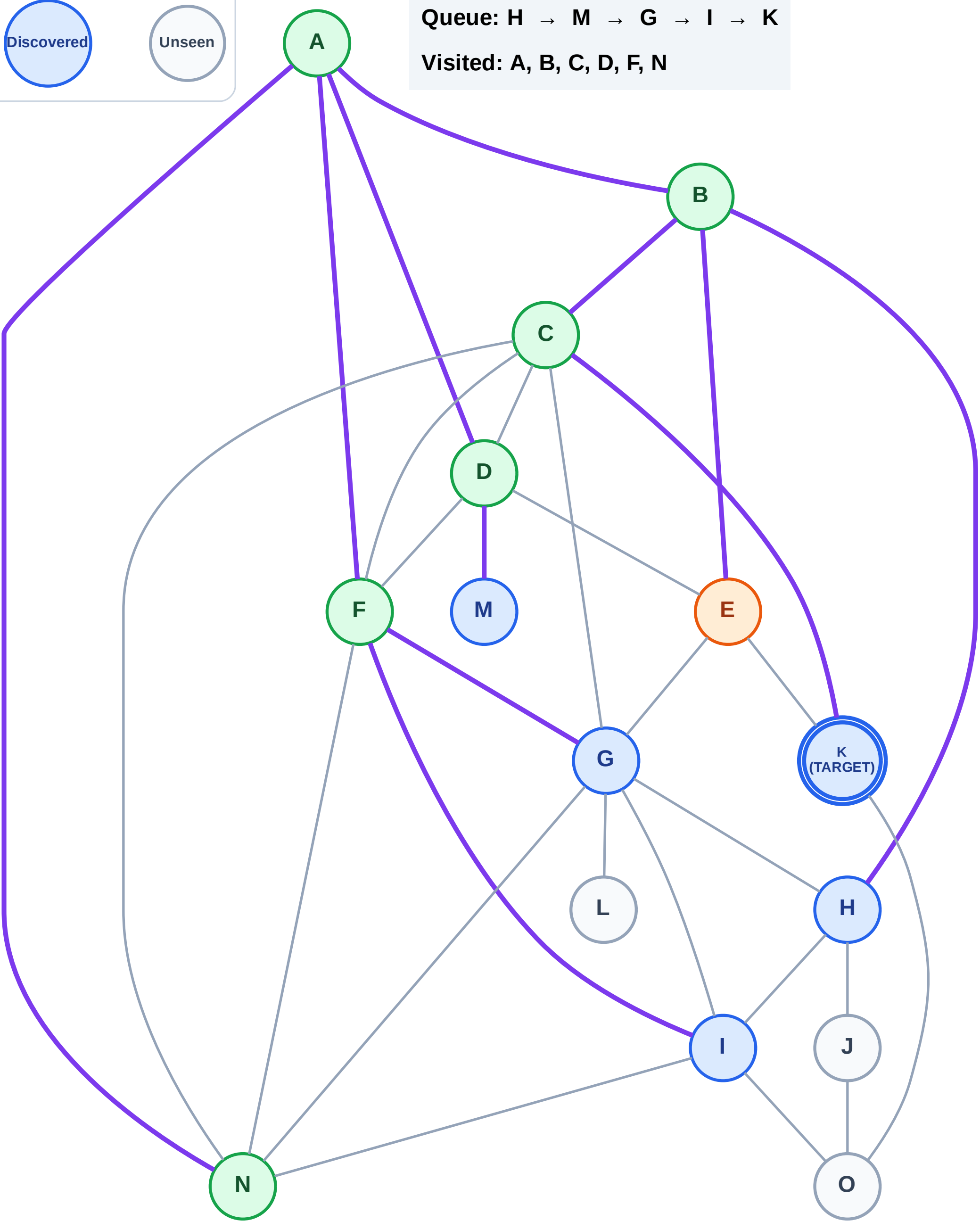
Step 14: Process node E

Legend



Queue: H → M → G → I → K

Visited: A, B, C, D, F, N



BFS Traversal

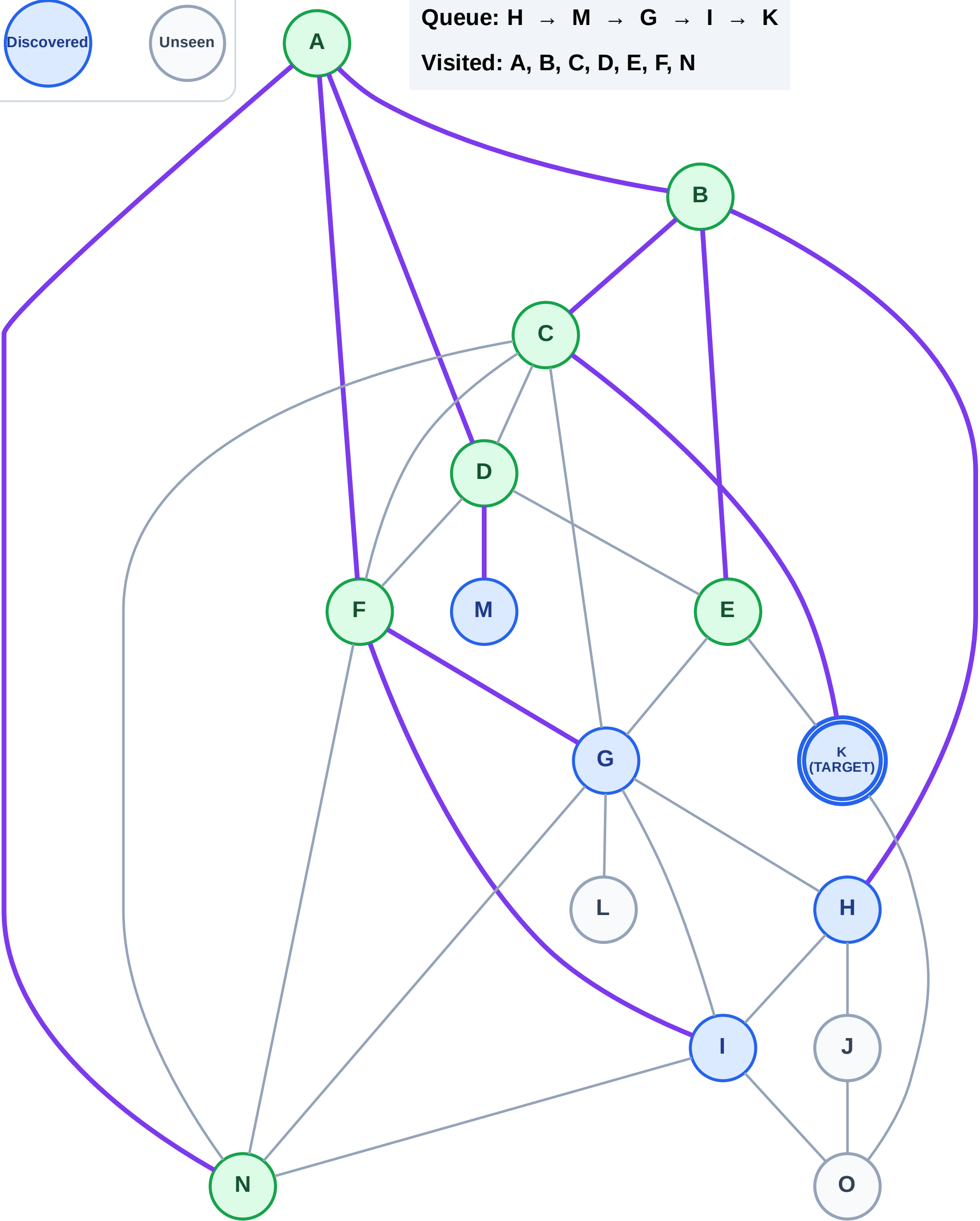
Step 15: No new neighbors from E

Legend



Queue: H → M → G → I → K

Visited: A, B, C, D, E, F, N



BFS Traversal

Step 16: Process node H

Legend

Complete

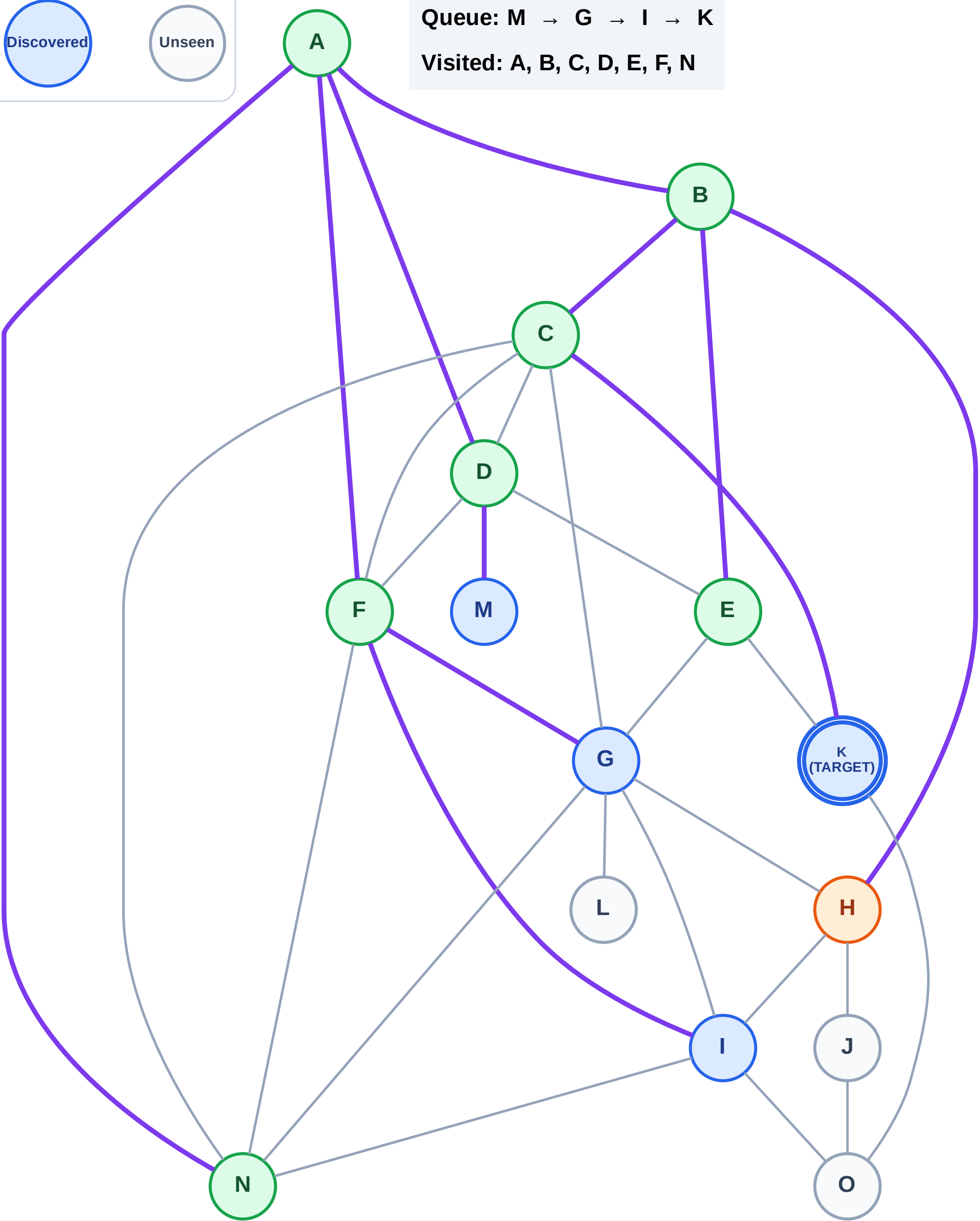
Processing

Discovered

Unseen

Queue: M → G → I → K

Visited: A, B, C, D, E, F, N



BFS Traversal

Step 17: Discover J from H

Legend

Complete

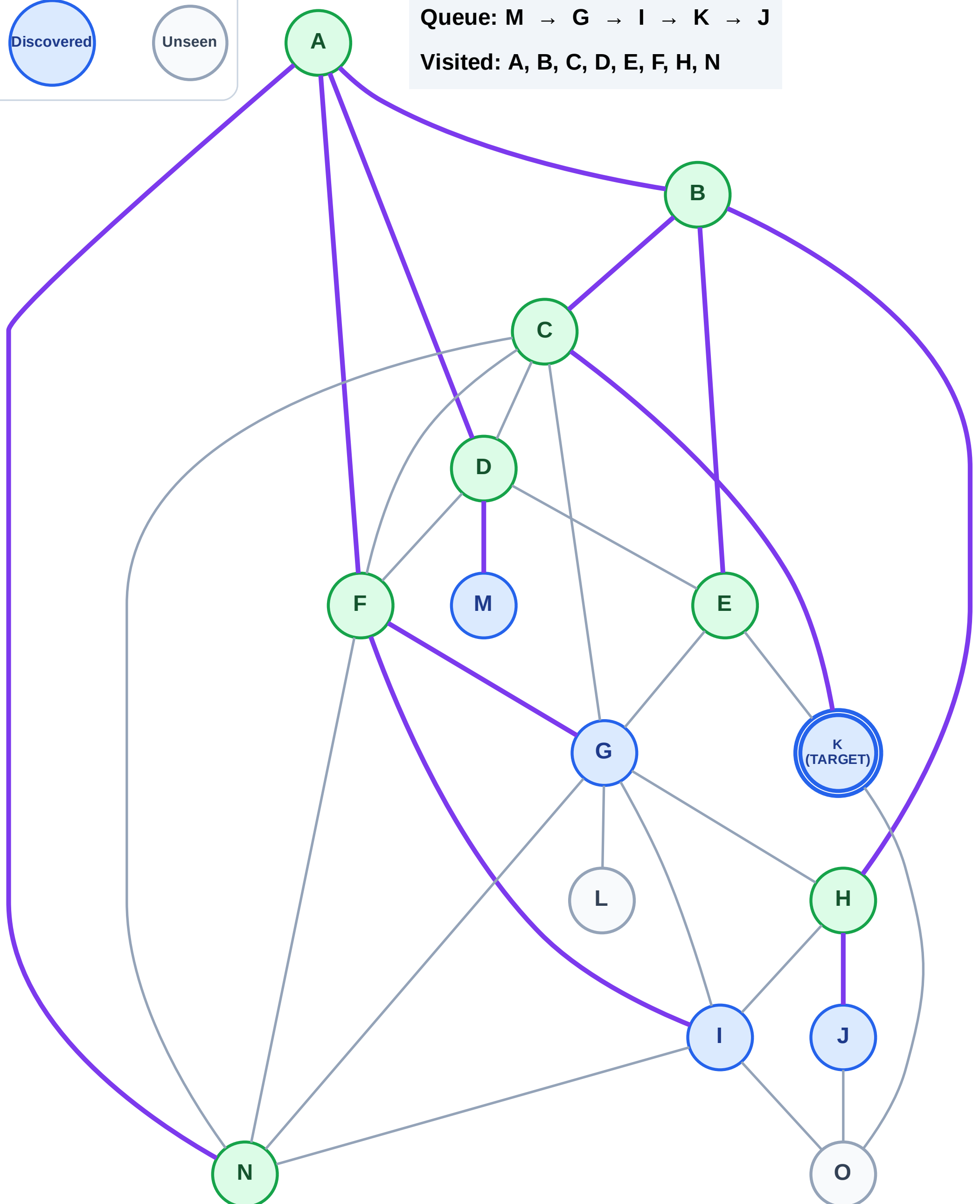
Processing

Discovered

Unseen

Queue: M → G → I → K → J

Visited: A, B, C, D, E, F, H, N



BFS Traversal

Step 18: Process node M

Legend

Complete

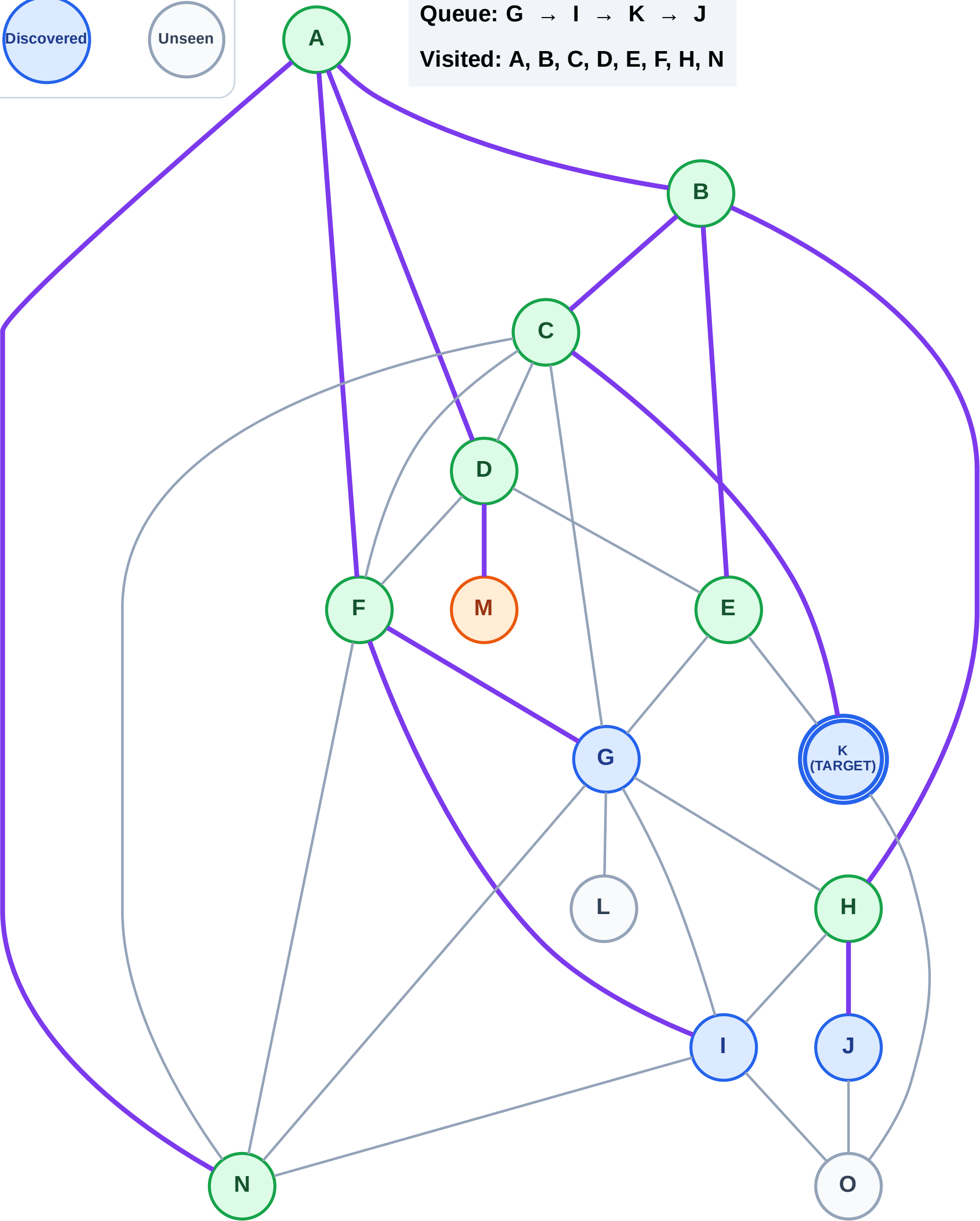
Processing

Discovered

Unseen

Queue: G → I → K → J

Visited: A, B, C, D, E, F, H, N



BFS Traversal

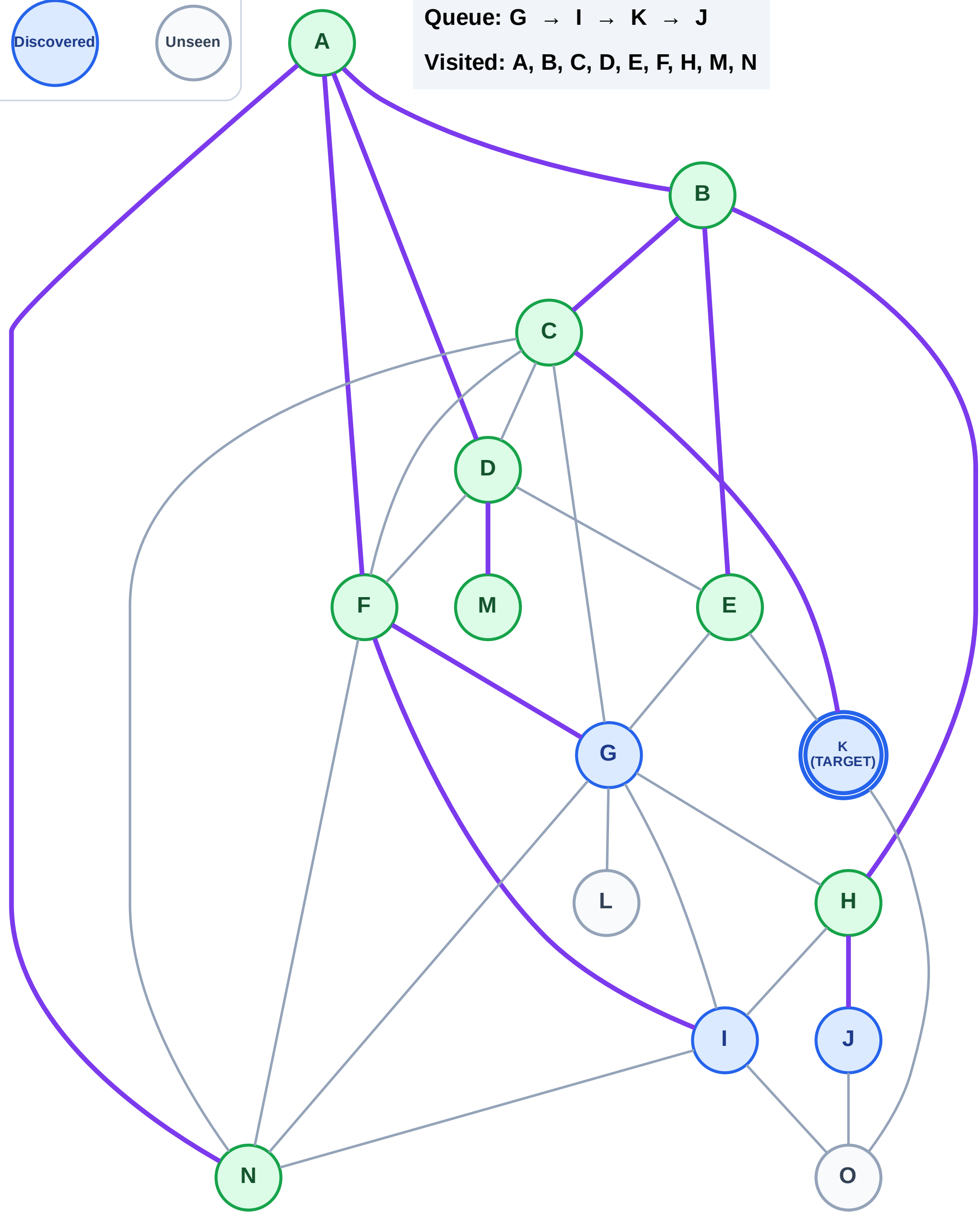
Step 19: No new neighbors from M

Legend



Queue: G → I → K → J

Visited: A, B, C, D, E, F, H, M, N



BFS Traversal

Step 20: Process node G

Legend

Complete

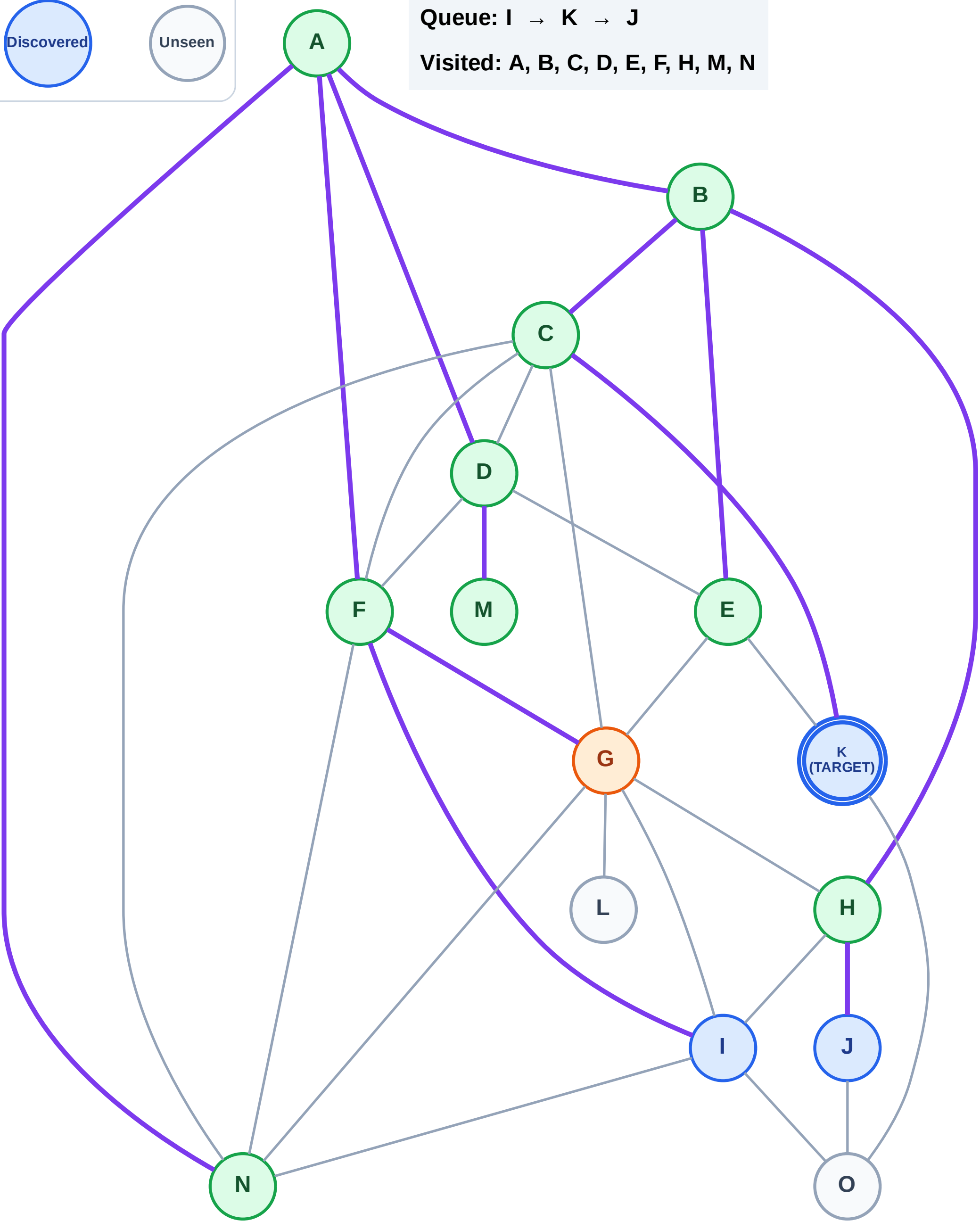
Processing

Discovered

Unseen

Queue: I → K → J

Visited: A, B, C, D, E, F, H, M, N



BFS Traversal

Step 21: Discover L from G

Legend

Complete

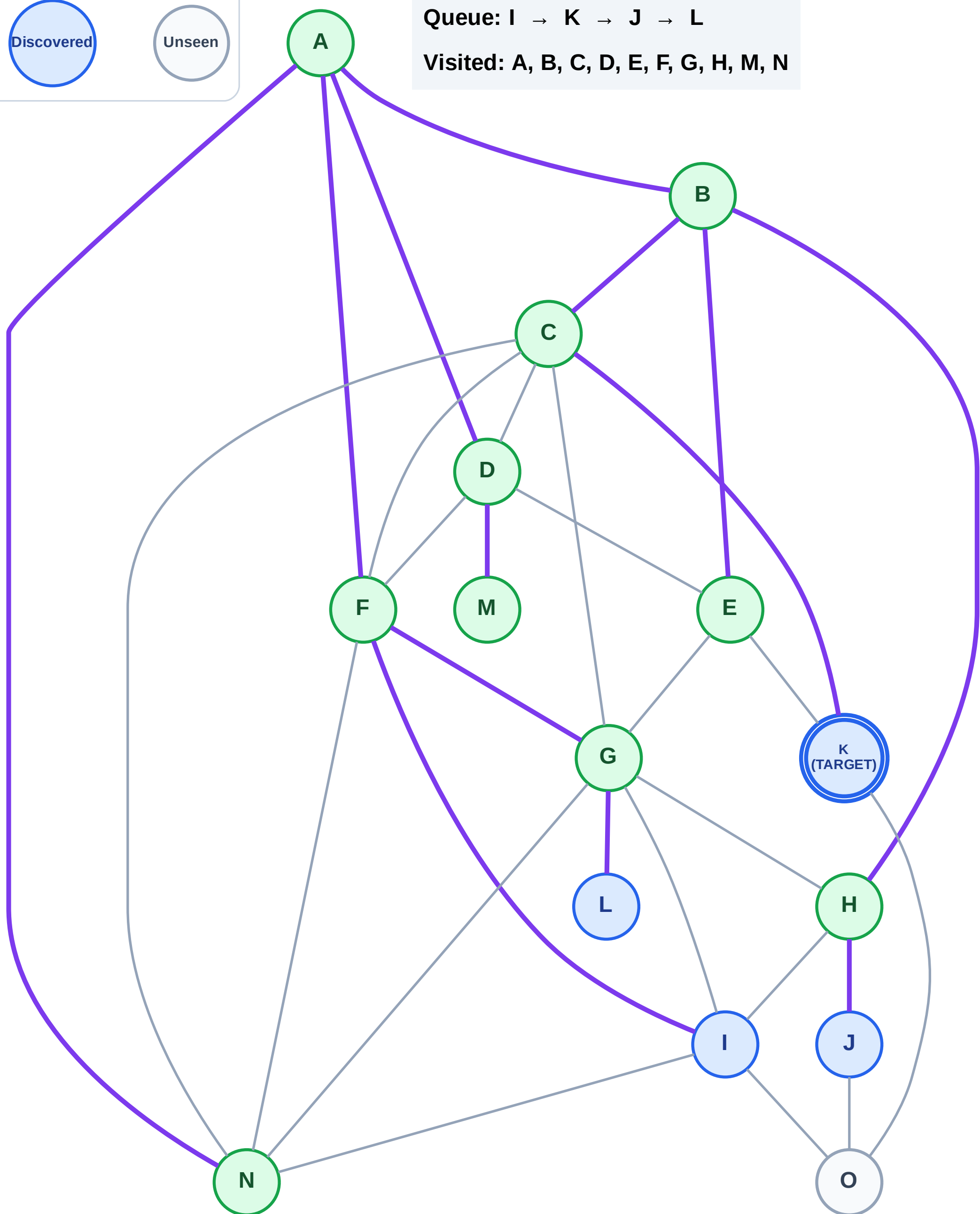
Processing

Discovered

Unseen

Queue: I → K → J → L

Visited: A, B, C, D, E, F, G, H, M, N



BFS Traversal

Step 22: Process node I

Legend

Complete

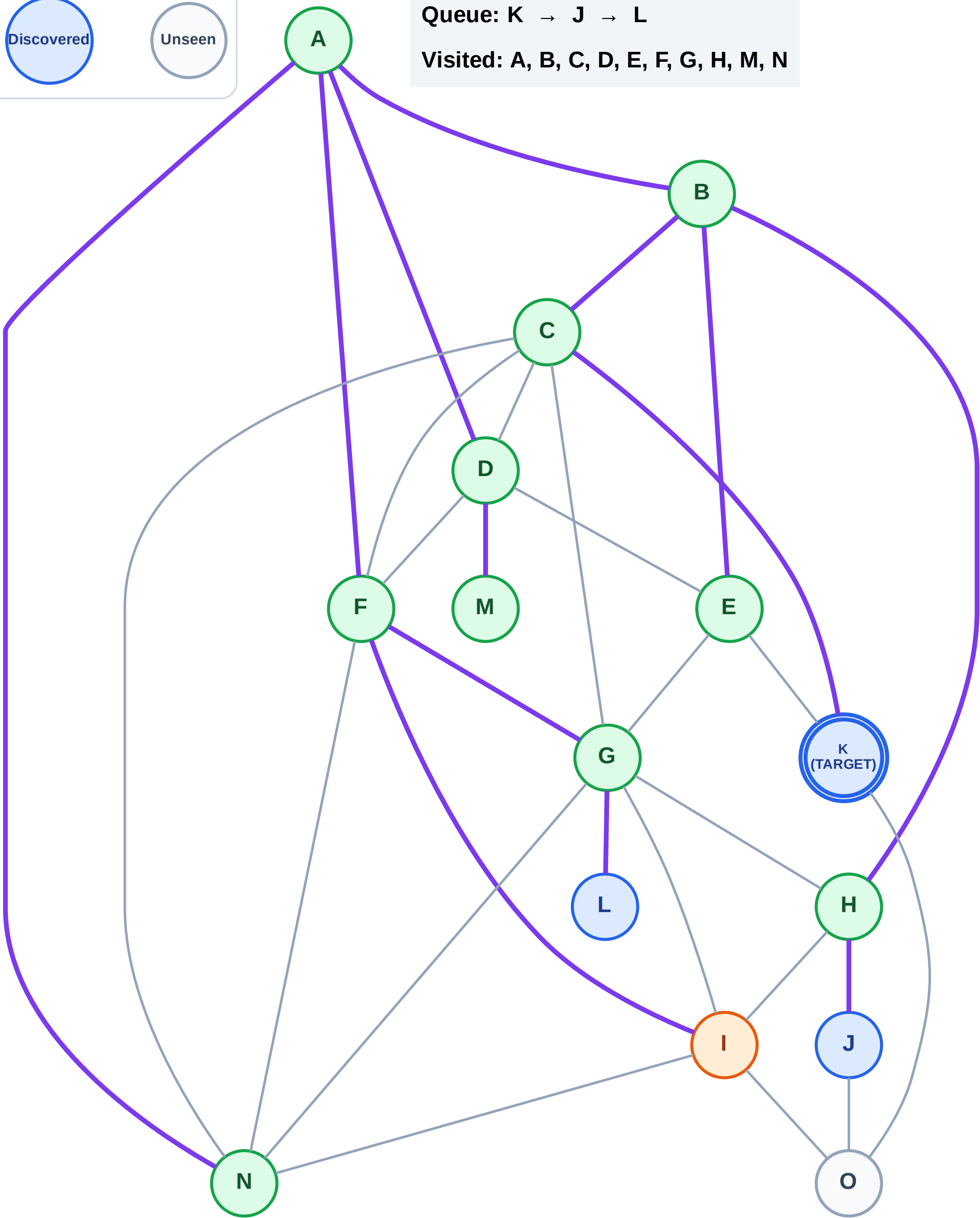
Processing

Discovered

Unseen

Queue: K → J → L

Visited: A, B, C, D, E, F, G, H, M, N



BFS Traversal

Step 23: Discover O from I

Legend

Complete

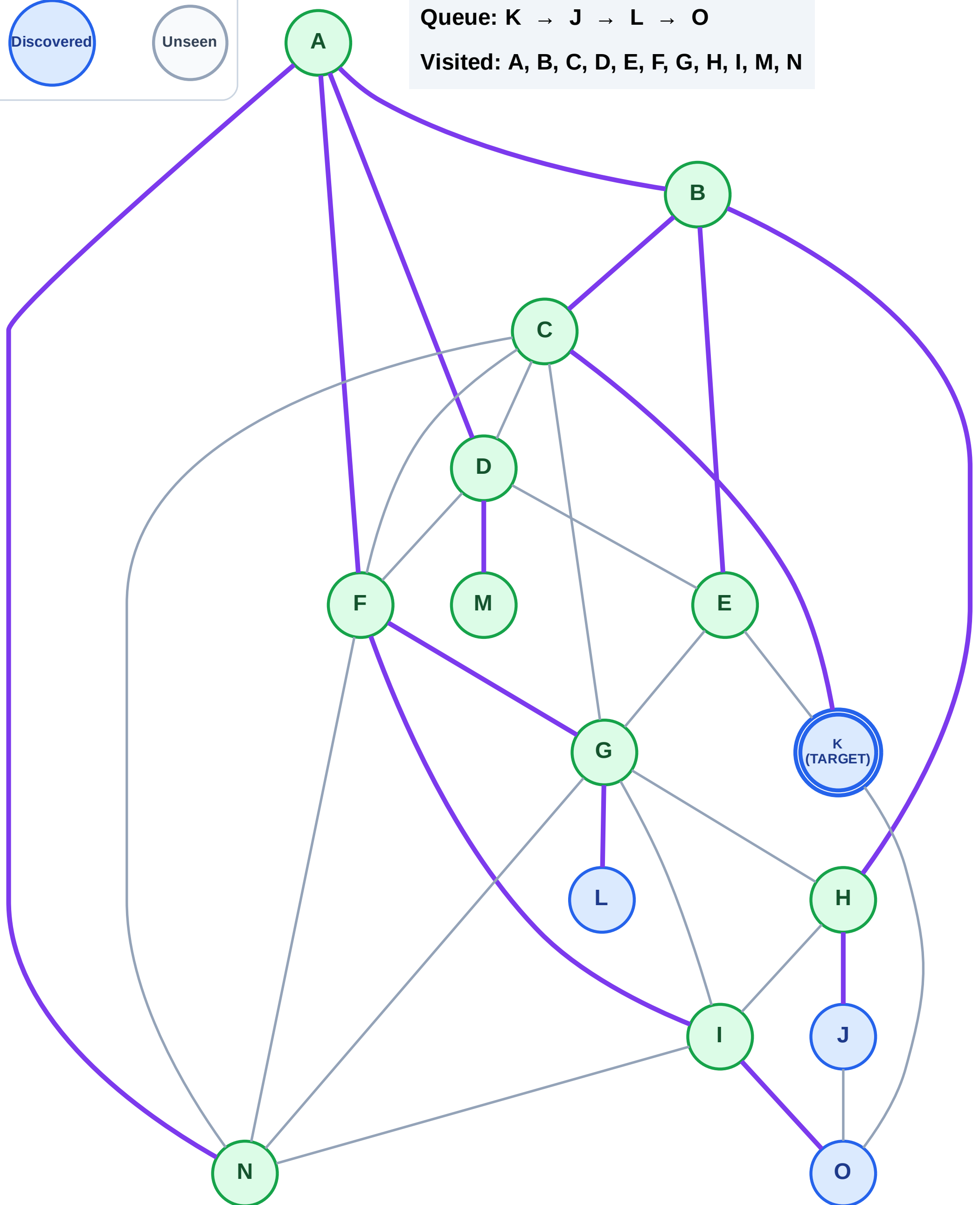
Processing

Discovered

Unseen

Queue: K → J → L → O

Visited: A, B, C, D, E, F, G, H, I, M, N



BFS Traversal

Step 24: Process node K

Legend

Complete

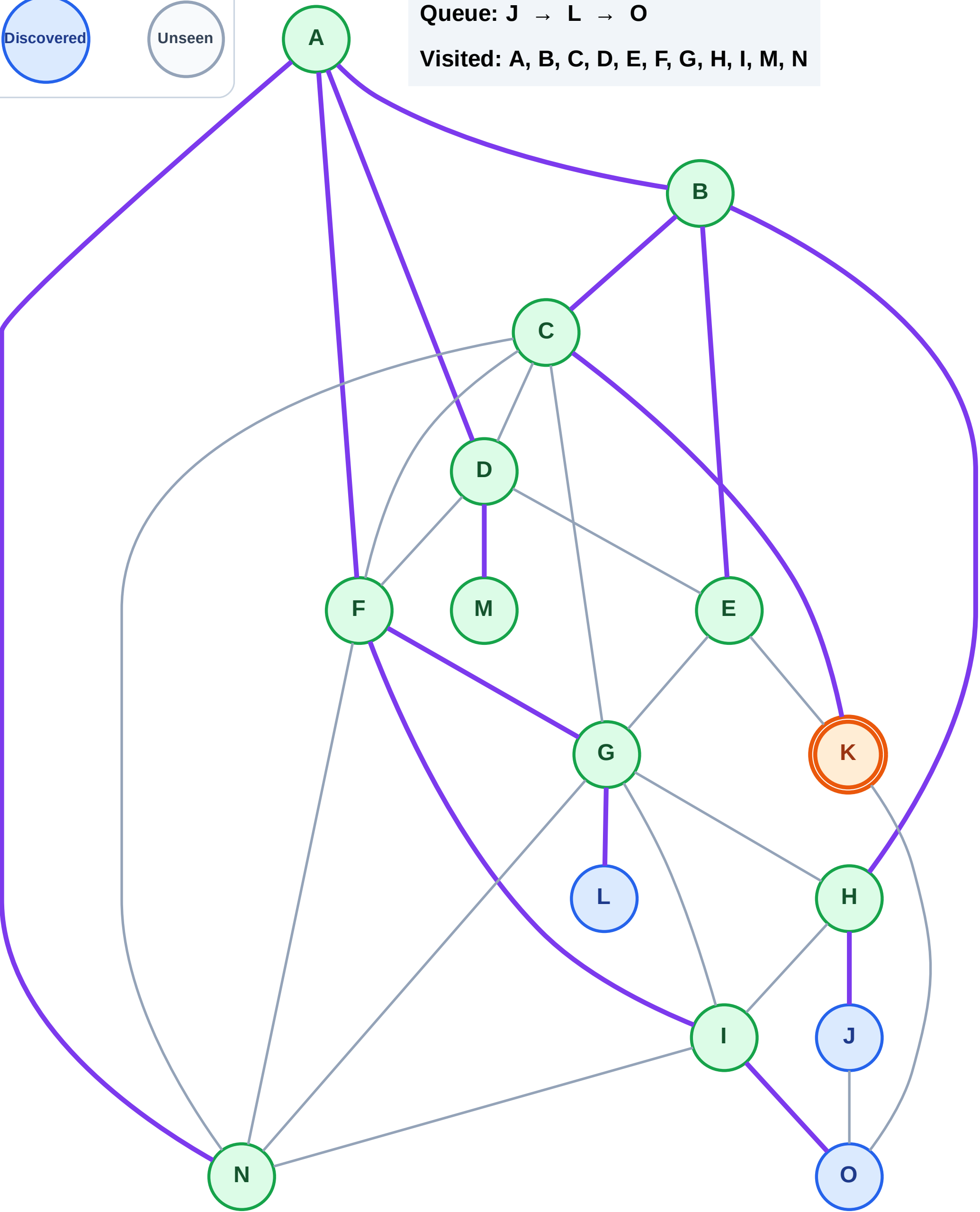
Processing

Discovered

Unseen

Queue: J → L → O

Visited: A, B, C, D, E, F, G, H, I, M, N



BFS Traversal

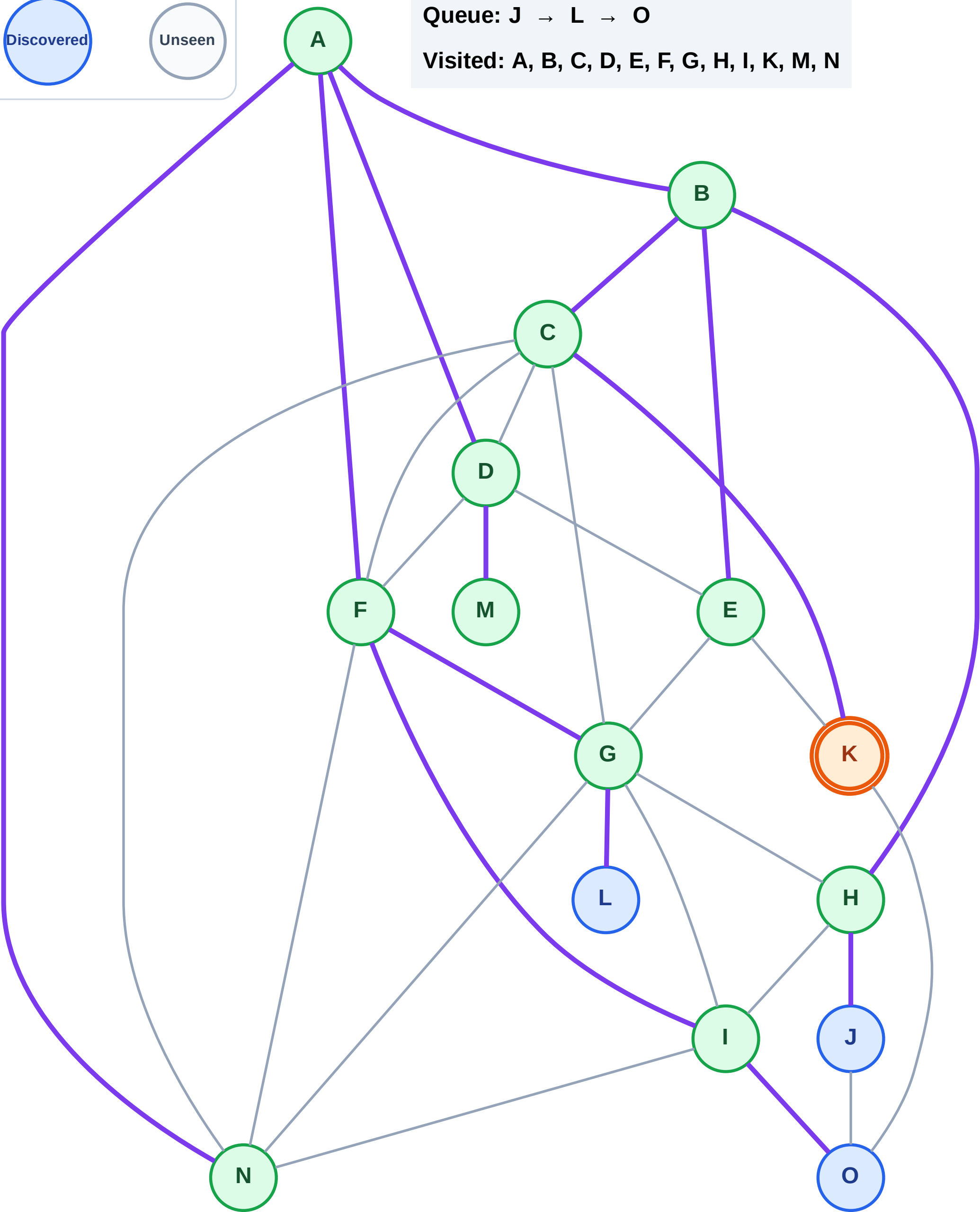
Step 25: TARGET NODE K REACHED! Path (3 hops): A → B → C → K

Legend



Queue: J → L → O

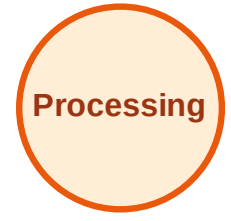
Visited: A, B, C, D, E, F, G, H, I, K, M, N



DFS Traversal

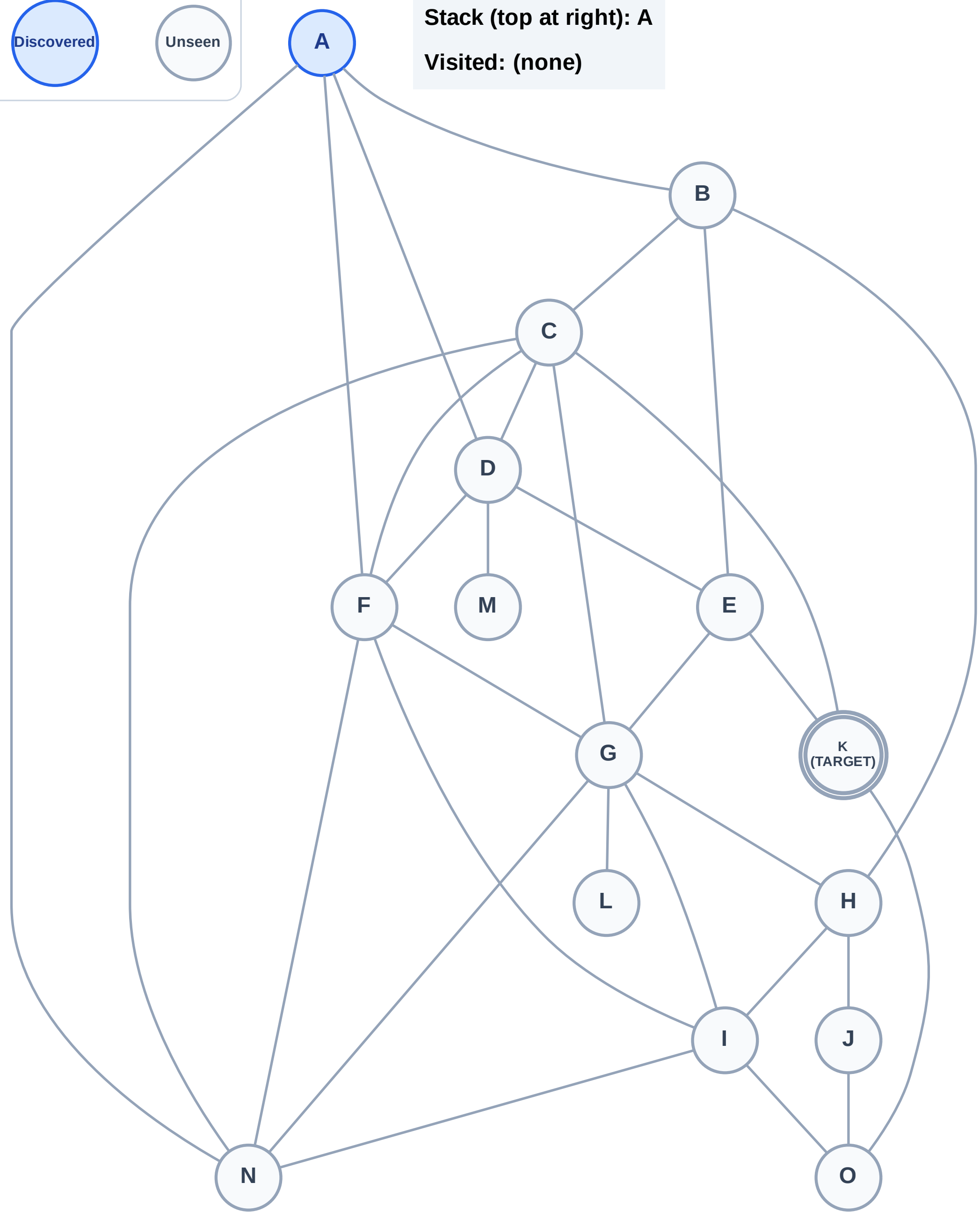
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

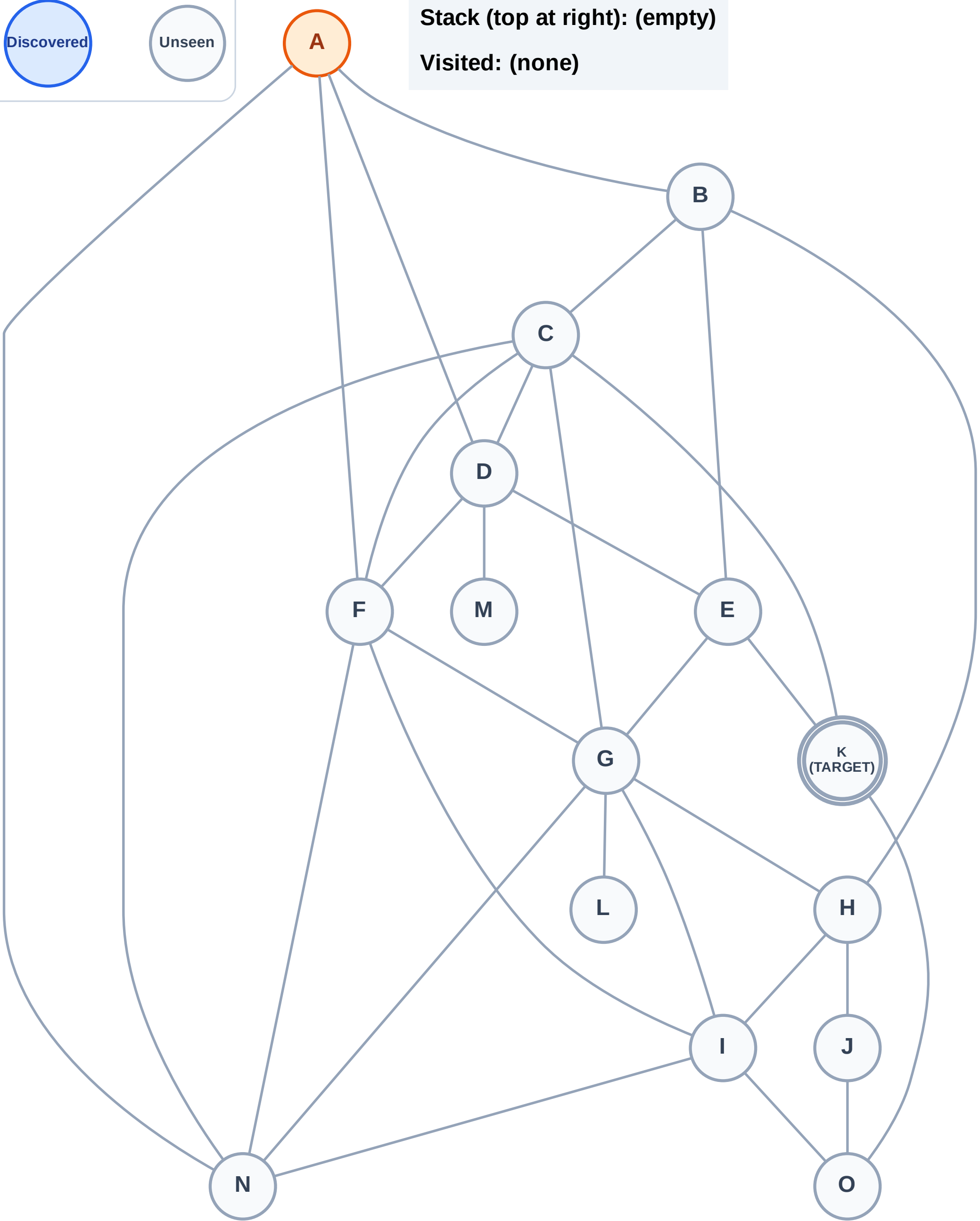
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



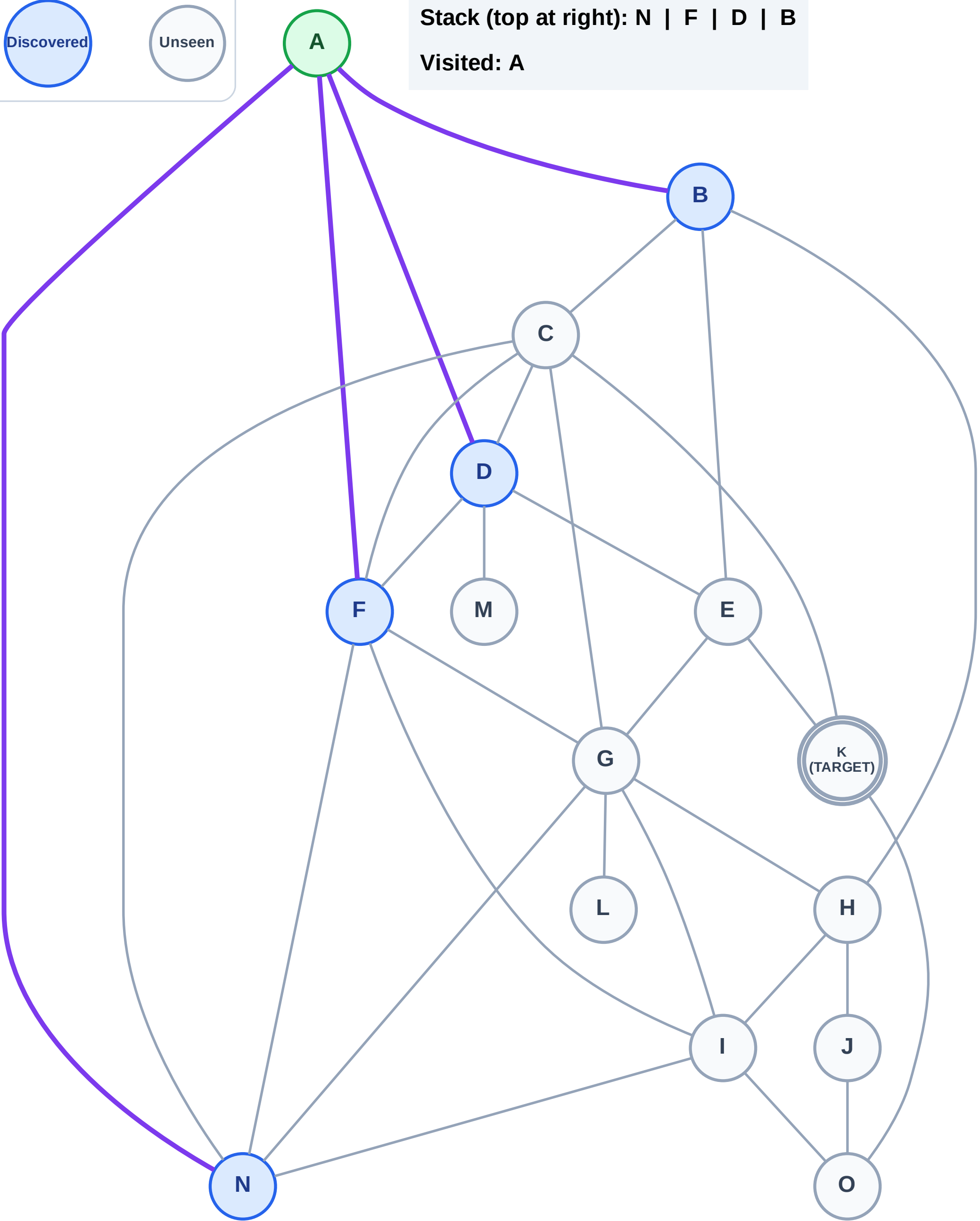
DFS Traversal

Step 3: Discover N, F, D, B from A

Legend



Stack (top at right): N | F | D | B
Visited: A



DFS Traversal

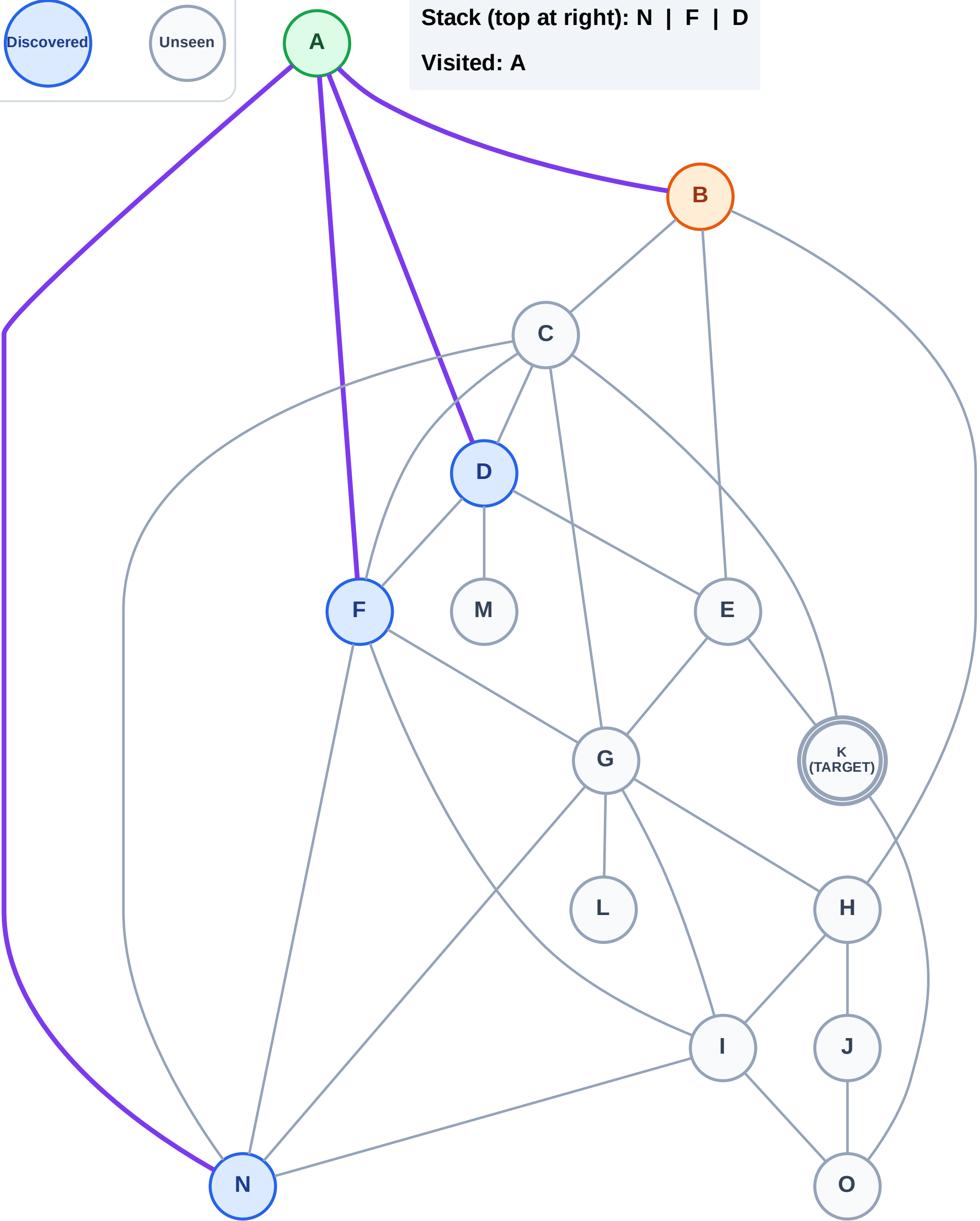
Step 4: Process node B

Legend



Stack (top at right): N | F | D

Visited: A



DFS Traversal

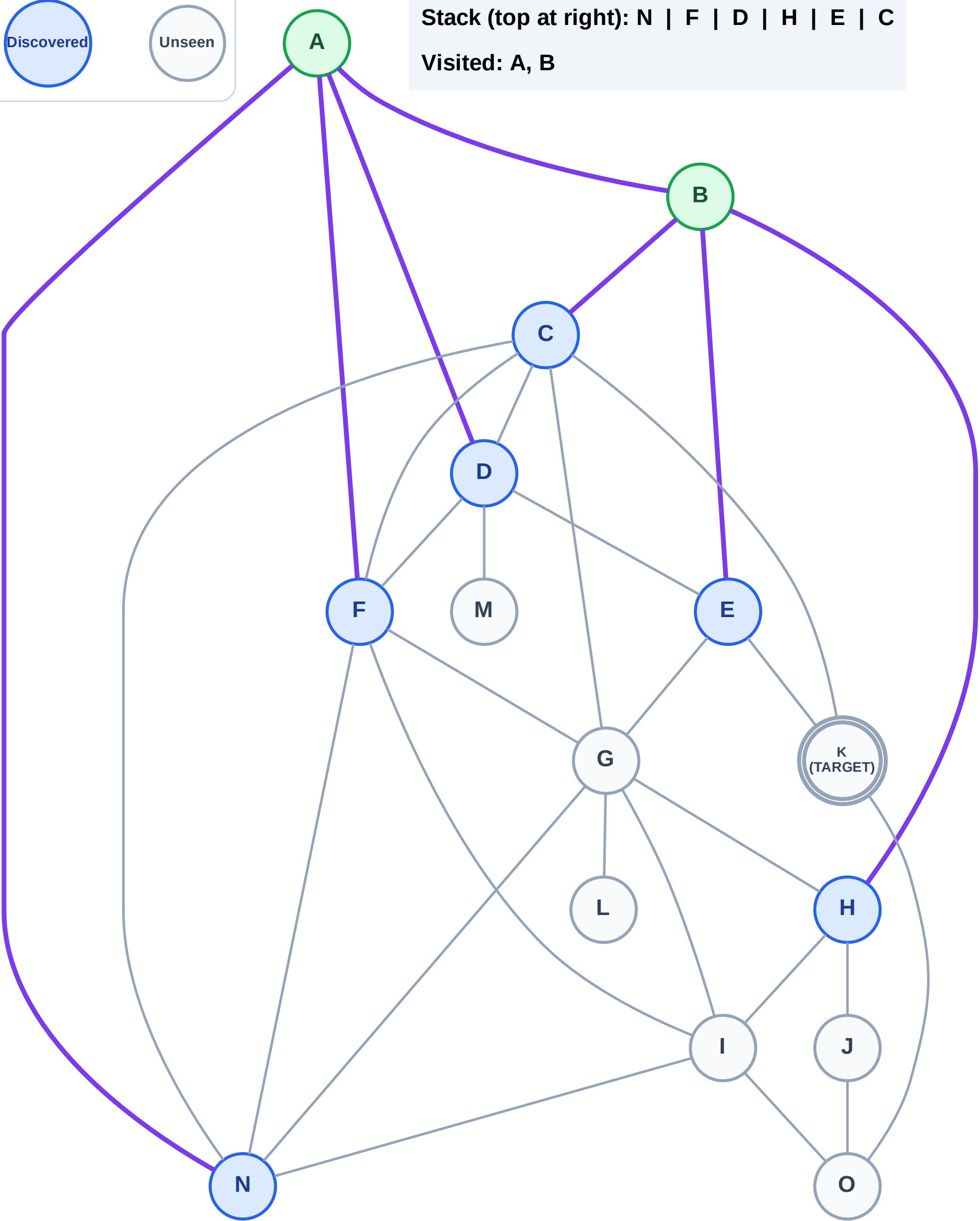
Step 5: Discover H, E, C from B

Legend



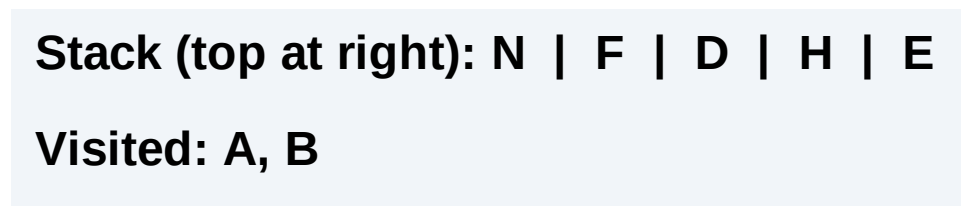
Stack (top at right): N | F | D | H | E | C

Visited: A, B



Step 6: Process node C

Step 6: Process node C



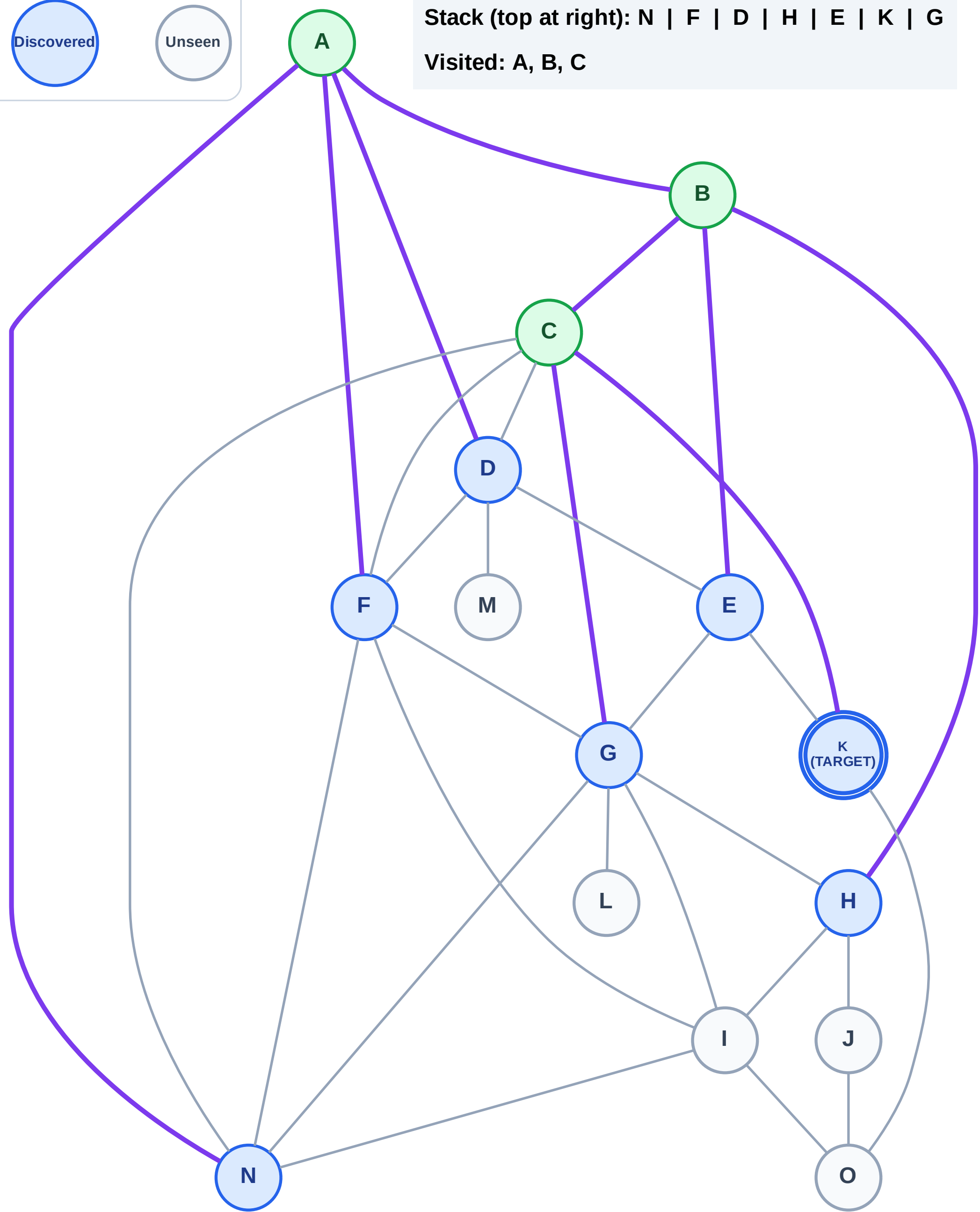
DFS Traversal

Step 7: Discover K, G from C

Legend



Stack (top at right): N | F | D | H | E | K | G
Visited: A, B, C



DFS Traversal

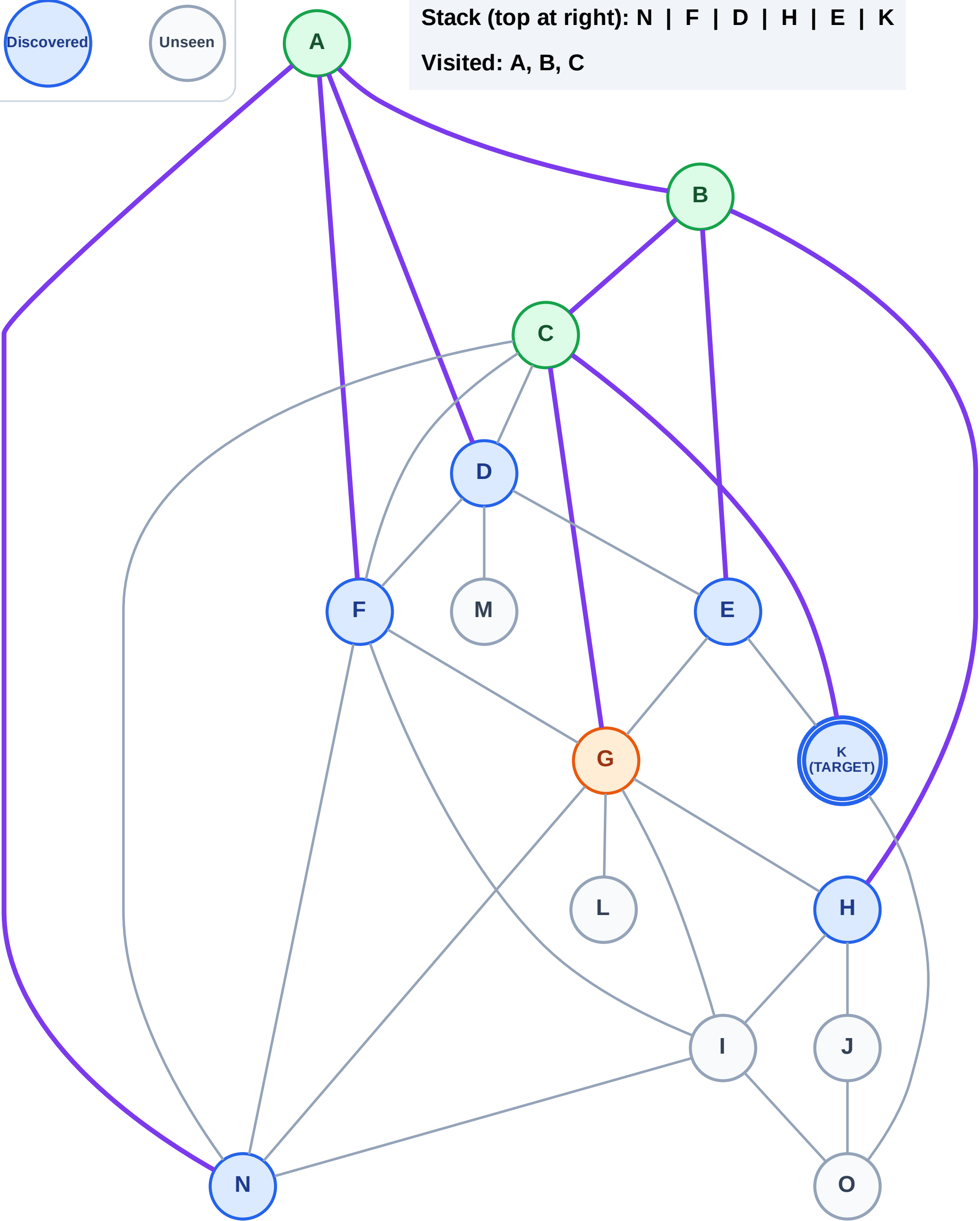
Step 8: Process node G

Legend



Stack (top at right): N | F | D | H | E | K

Visited: A, B, C



DFS Traversal

Step 9: Discover L, I from G

Legend

Complete

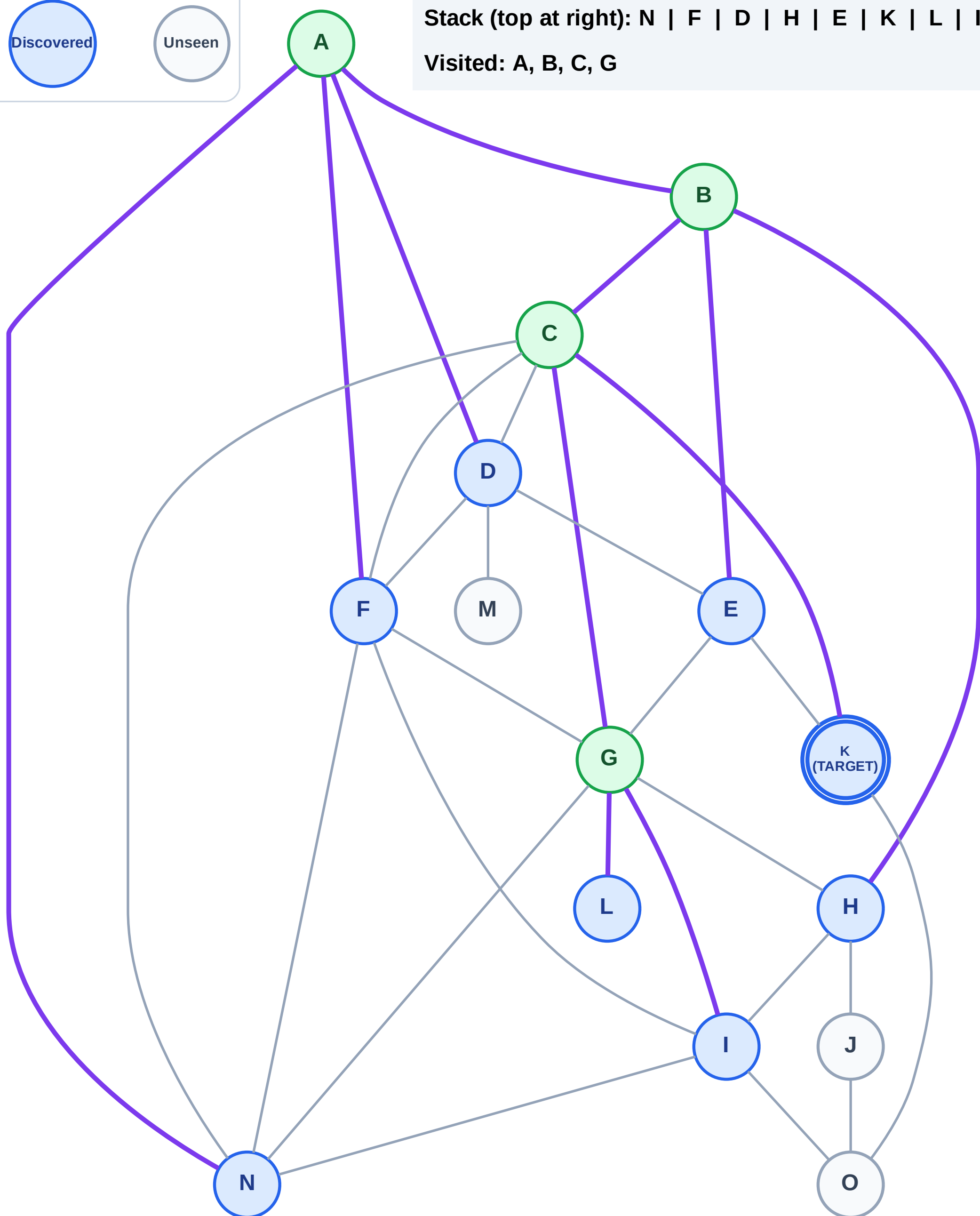
Processing

Discovered

Unseen

Stack (top at right): N | F | D | H | E | K | L | I

Visited: A, B, C, G



DFS Traversal

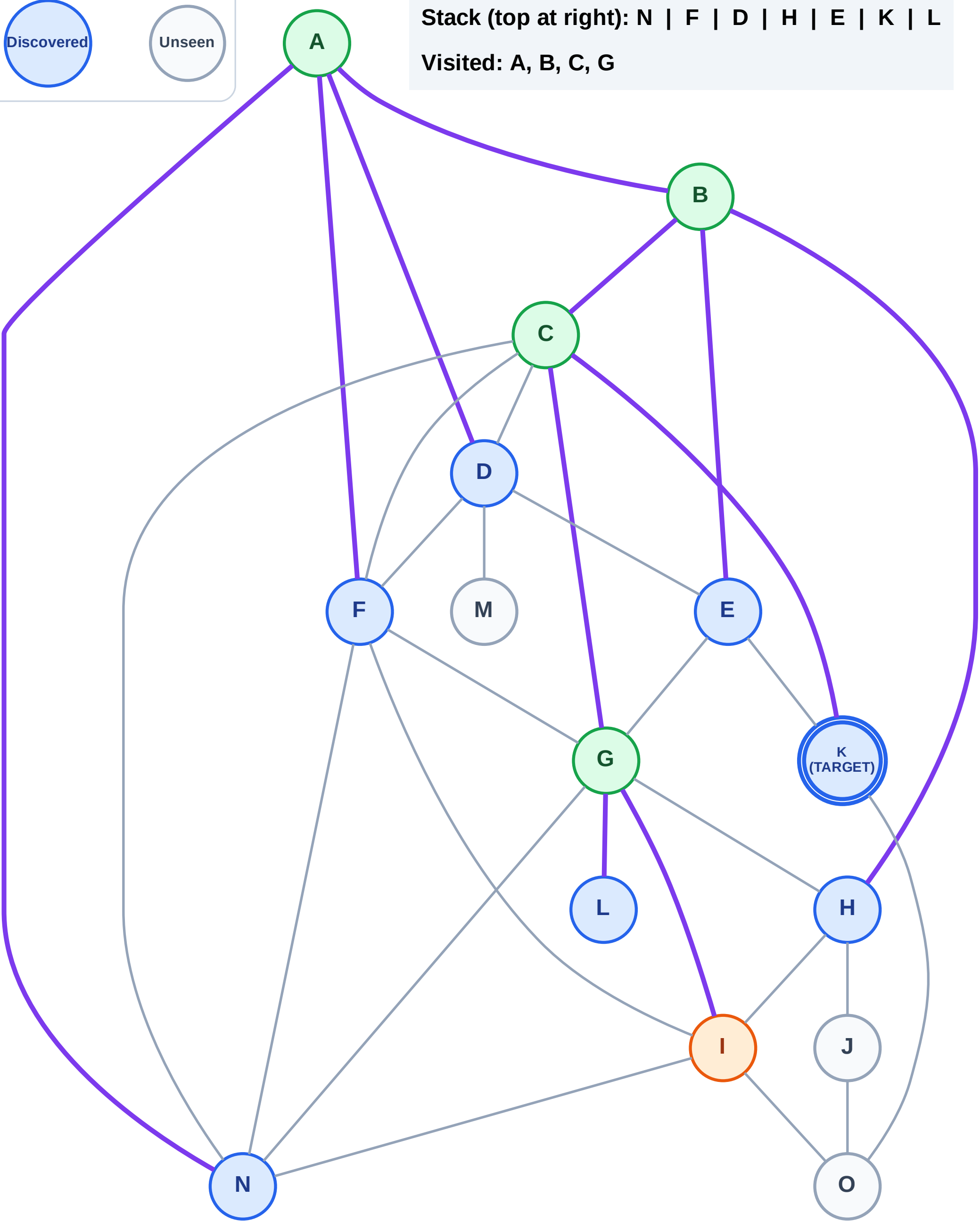
Step 10: Process node I

Legend



Stack (top at right): N | F | D | H | E | K | L

Visited: A, B, C, G



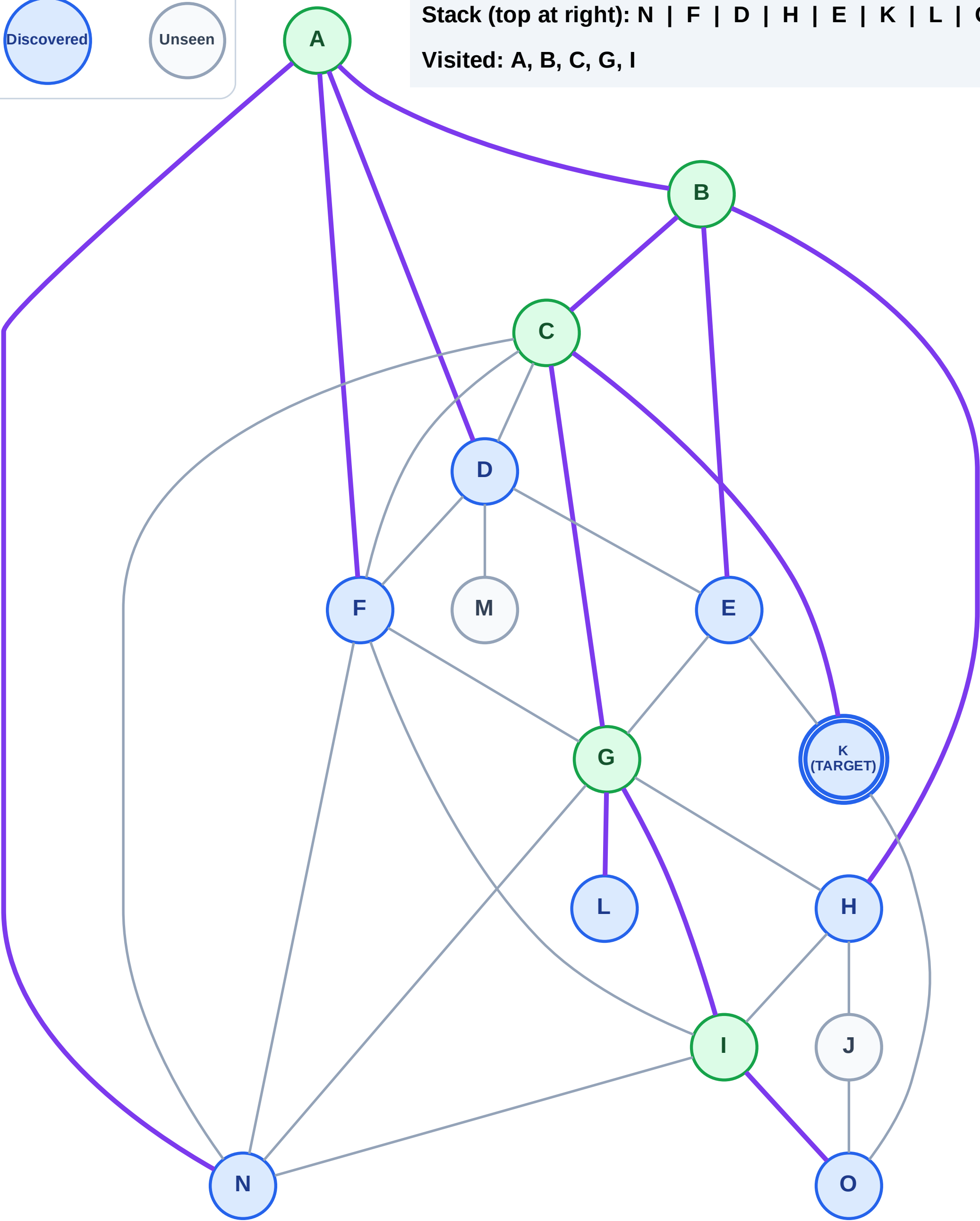
Step 11: Discover O from



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DFS Traversal

Step 12: Process node O

Legend

Complete

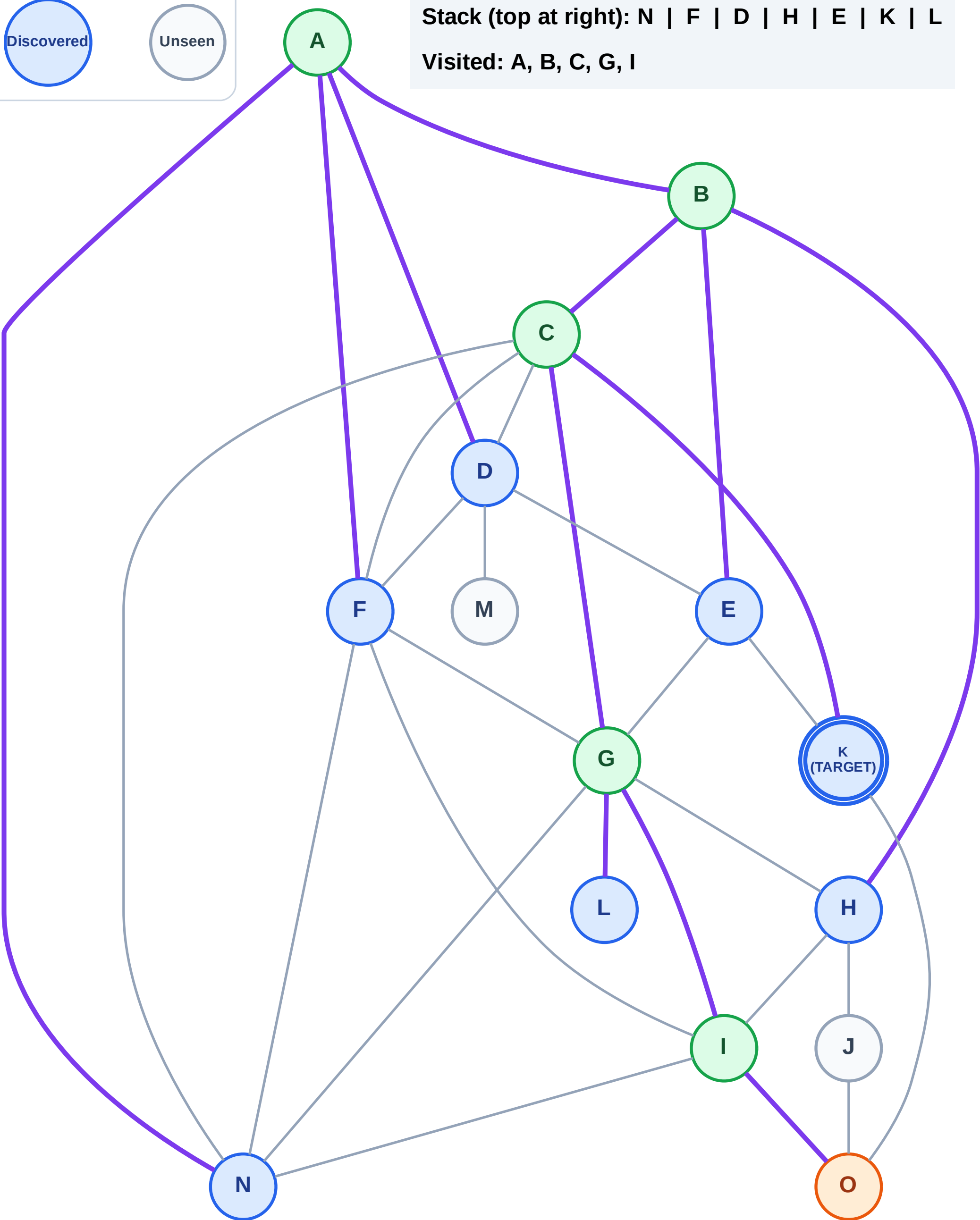
Processing

Discovered

Unseen

Stack (top at right): N | F | D | H | E | K | L

Visited: A, B, C, G, I



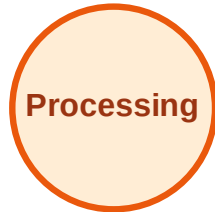
DFS Traversal

Step 13: Discover J from O

Legend



Complete



Processing



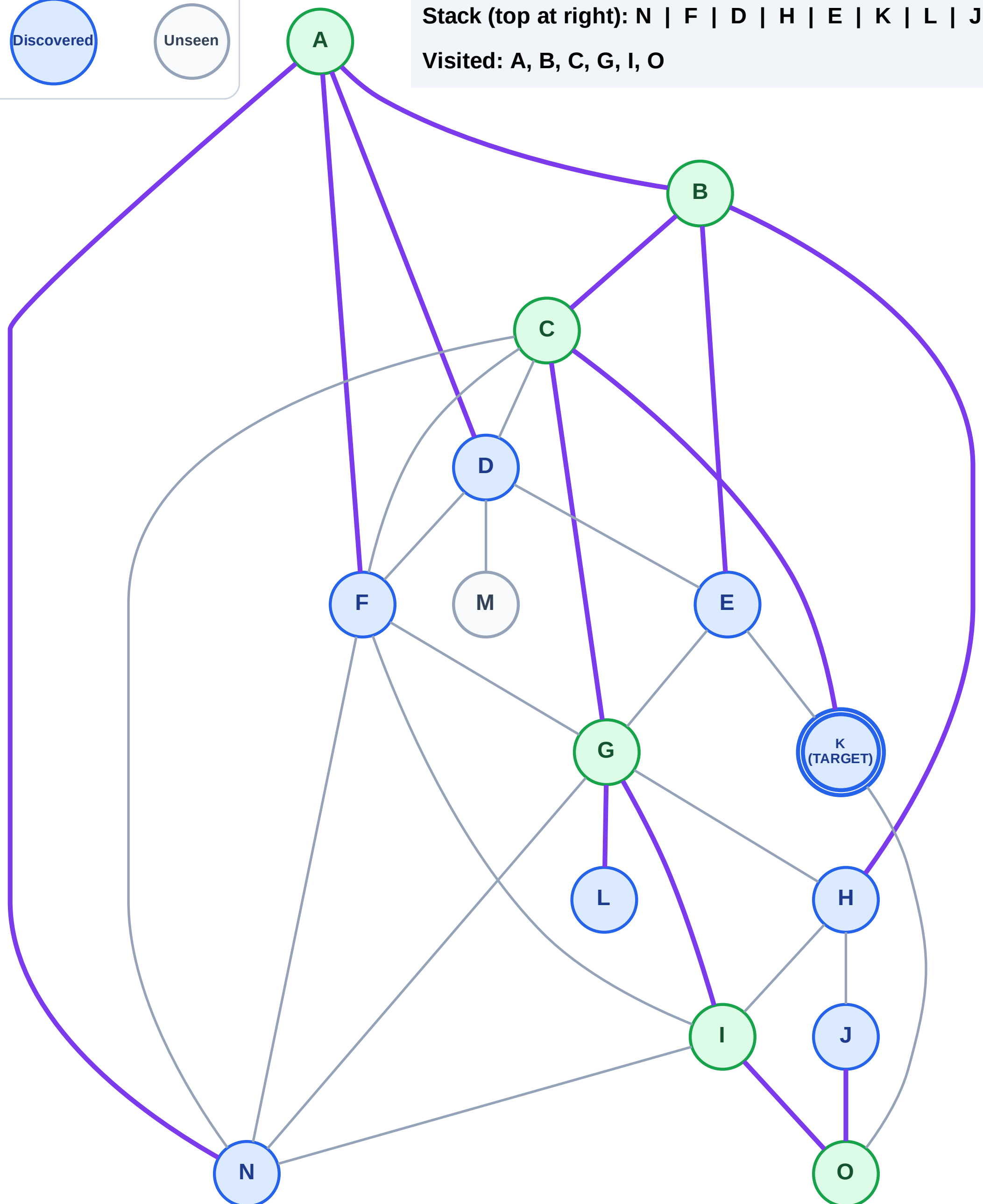
Discovered



Unseen

Stack (top at right): N | F | D | H | E | K | L | J

Visited: A, B, C, G, I, O



DFS Traversal

Step 14: Process node J

Legend

Complete

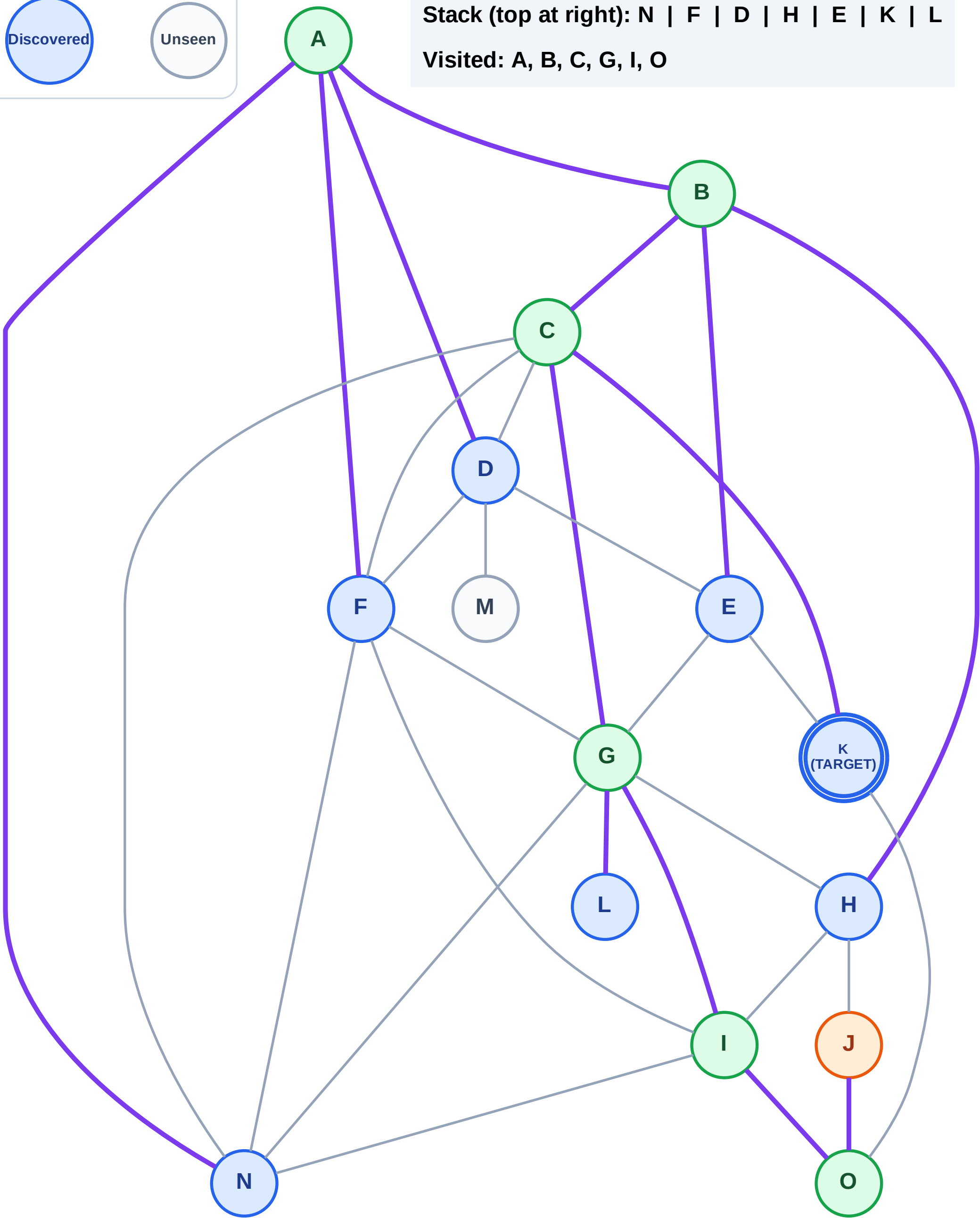
Processing

Discovered

Unseen

Stack (top at right): N | F | D | H | E | K | L

Visited: A, B, C, G, I, O



DFS Traversal

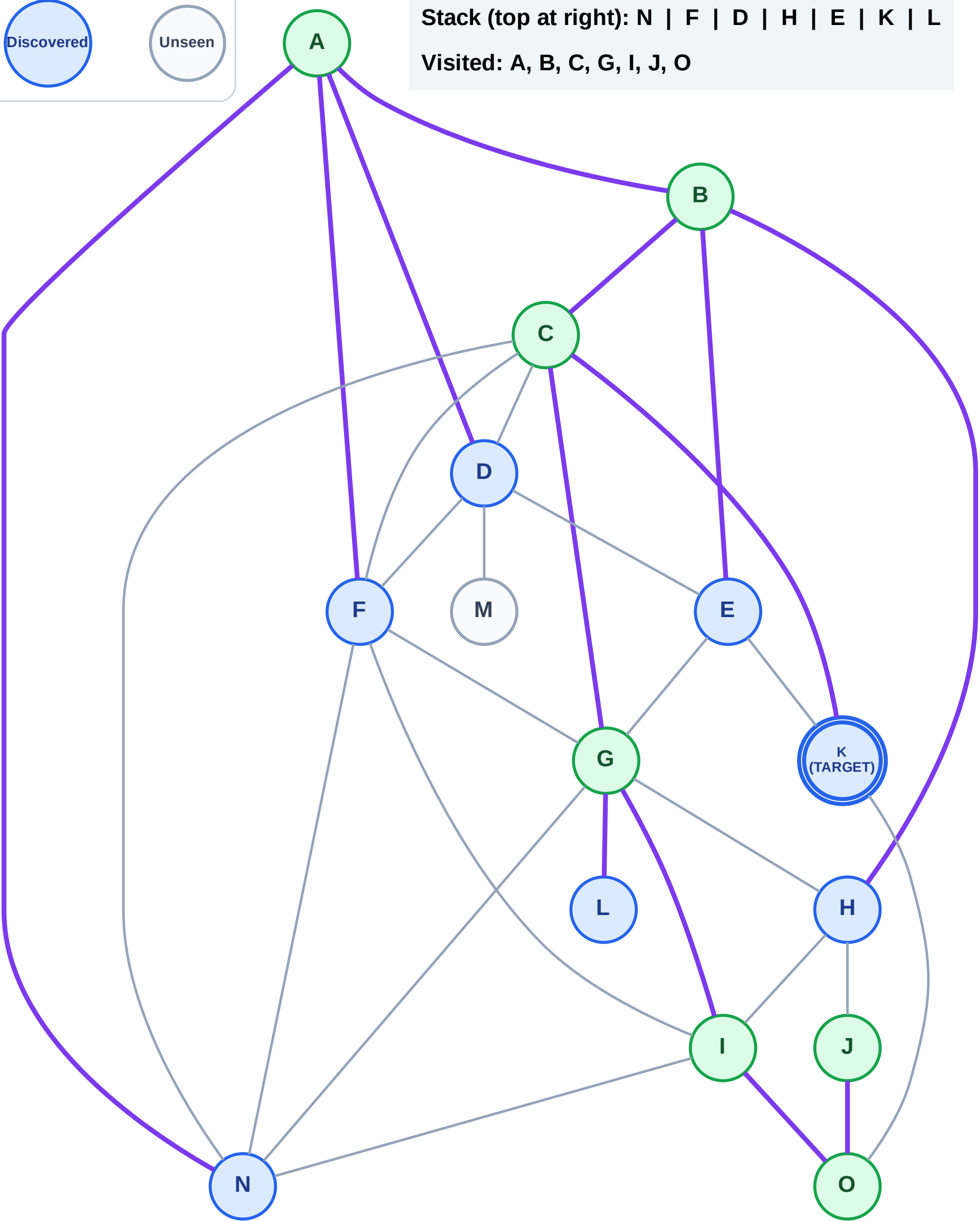
Step 15: No new neighbors from J

Legend



Stack (top at right): N | F | D | H | E | K | L

Visited: A, B, C, G, I, J, O



DFS Traversal

Step 16: Process node L

Legend

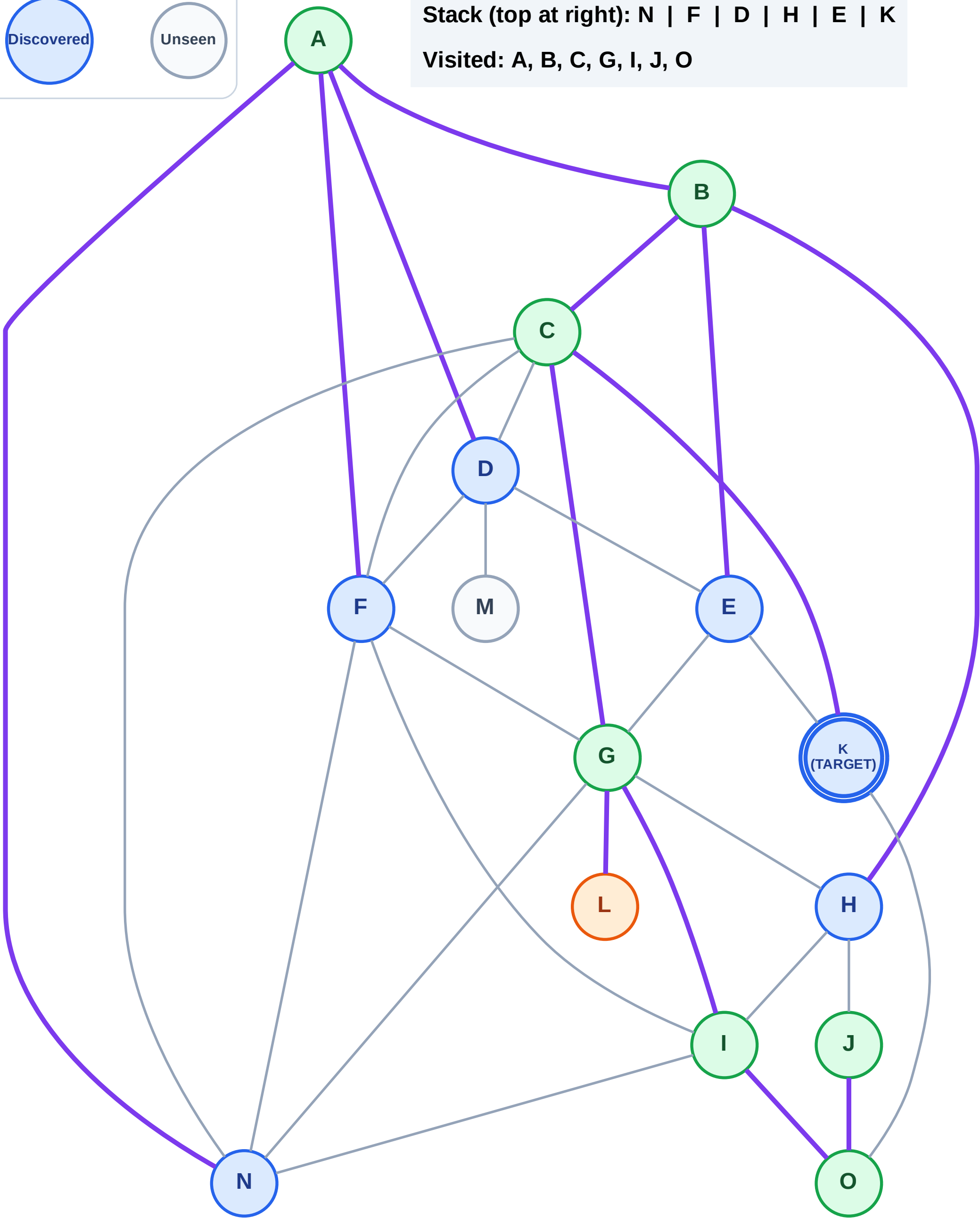
Complete

Processing

Discovered

Unseen

Stack (top at right): N | F | D | H | E | K
Visited: A, B, C, G, I, J, O



DFS Traversal

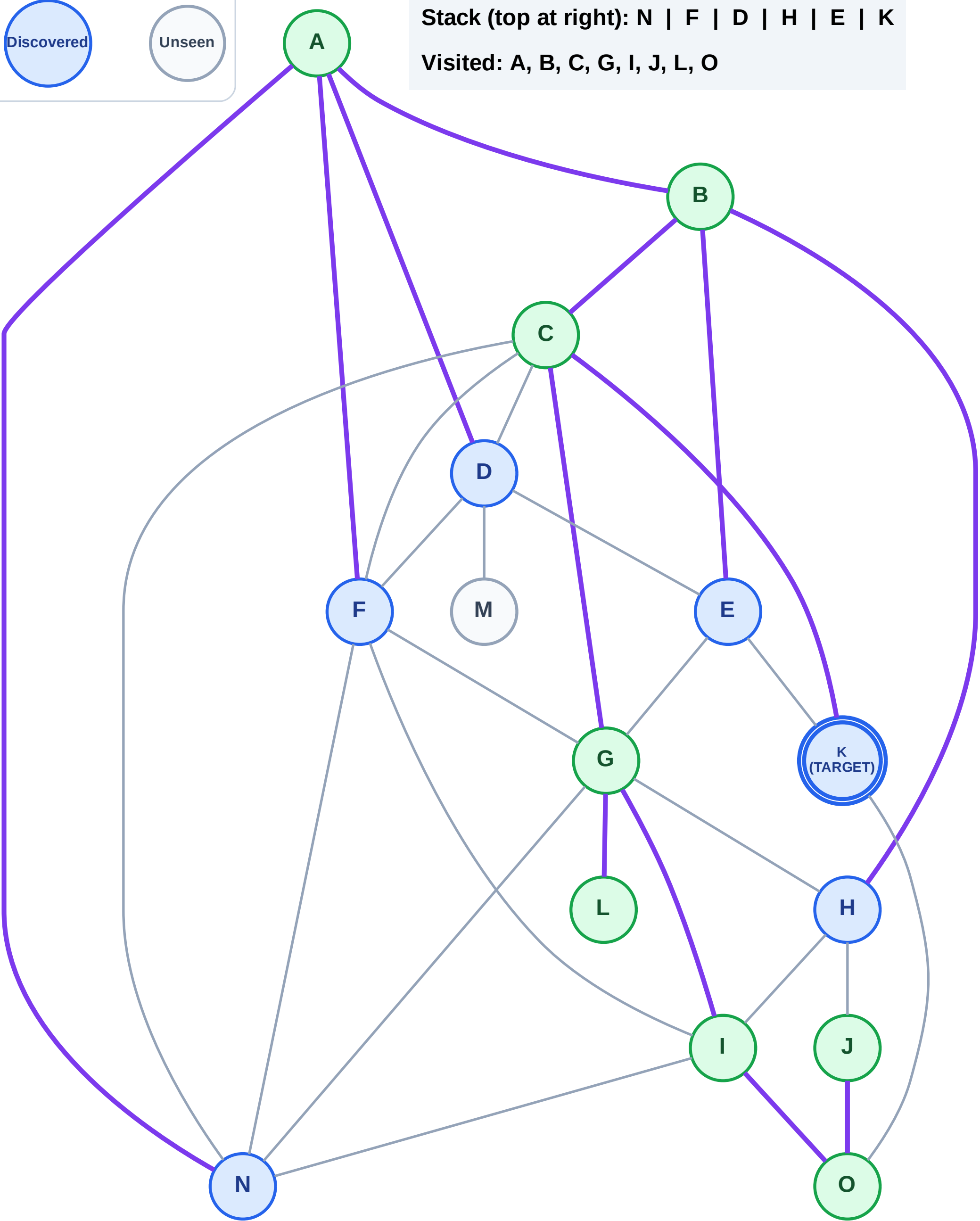
Step 17: No new neighbors from L

Legend



Stack (top at right): N | F | D | H | E | K

Visited: A, B, C, G, I, J, L, O



DFS Traversal

Step 18: Process node K

Legend

Complete

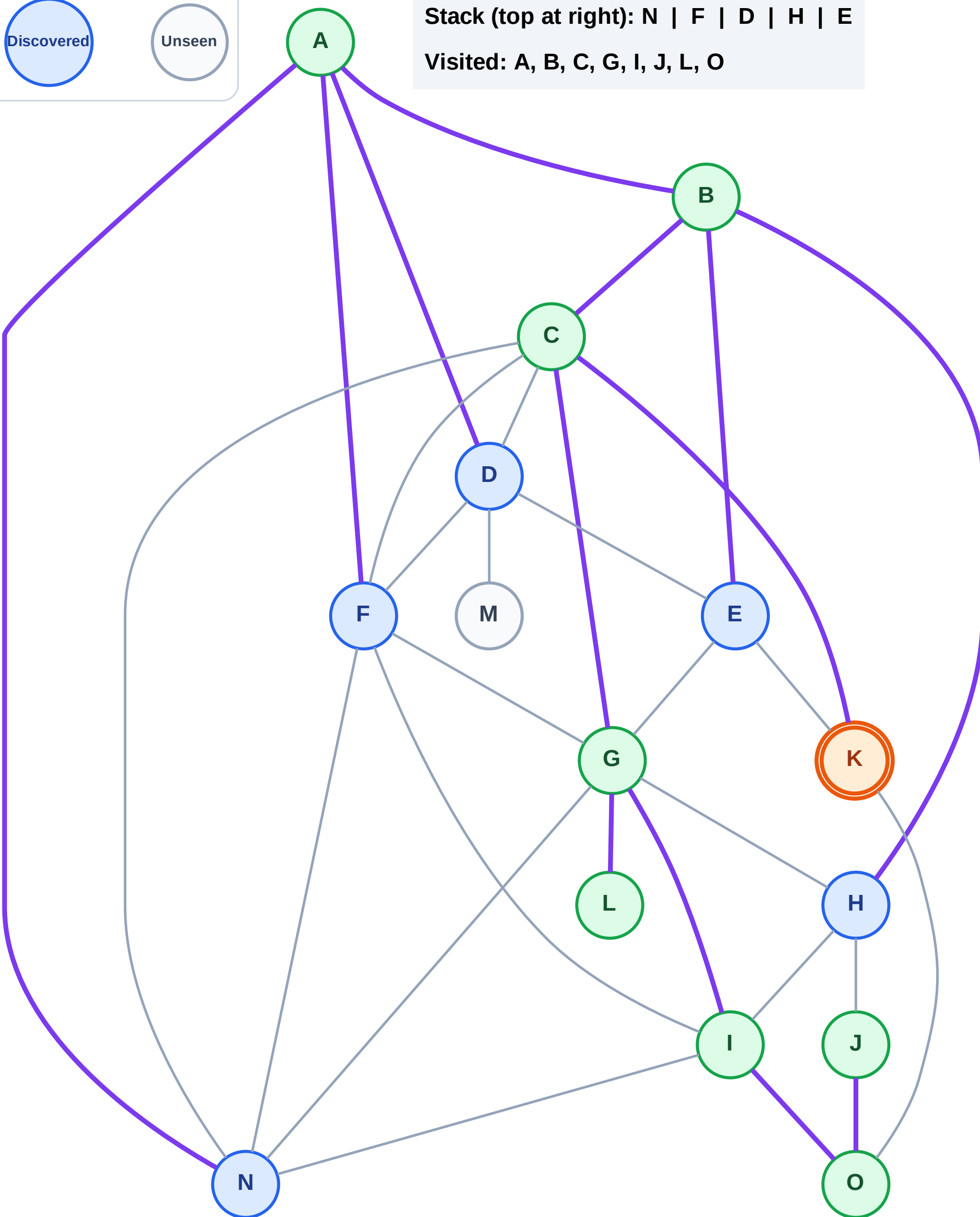
Processing

Discovered

Unseen

Stack (top at right): N | F | D | H | E

Visited: A, B, C, G, I, J, L, O



DFS Traversal

Step 19: TARGET NODE K REACHED! Path (3 hops): A → B → C → K

Legend



Stack (top at right): N | F | D | H | E

Visited: A, B, C, G, I, J, K, L, O

